Artificial Intelligence In Medicine 178 (2026) 103422

Contents lists available at ScienceDirect

Artificial Intelligence In Medicine

journal homepage: www.elsevier.com/locate/artmed

Research paper

Centralized pooling and federated learning for Canadian patient-level data sharing in multicenter medical AI: A scoping review

Omid Jafarinezhad [a,*], Qian Zhang [a], Ryan Rezai [b], Mohammad Noaeen [a], Aviv Shachak [a,c], Behrouz Far [d], Zahra Shakeri [a,c,e]

[a] Institute of Health Policy, Management and Evaluation, Dalla Lana School of Public Health, University of Toronto, Toronto, Canada
[b] Department of Electrical and Computer Engineering, McGill University, Montréal, Canada
[c] Faculty of Information, University of Toronto, Toronto, Canada
[d] Department of Electrical and Software Engineering, University of Calgary, Calgary, Canada
[e] Schwartz Reisman Institute, University of Toronto, Toronto, Canada

ARTICLE INFO

Dataset link: https://github.com/HIVE-UofT/Collaborative_Research_Review

Keywords:
Precision medicine
Health AI
Patient-level data sharing
Privacy-preserving machine learning
Federated and distributed learning

ABSTRACT

Algorithms that support screening, triage, and treatment decisions depend on training data drawn from patient populations. Limited access to patient-level records across institutions and jurisdictions can reduce representation and contribute to uneven model performance across populations. Canada's federated health system, where provinces and territories manage separate datasets and privacy regimes, limits multicenter medical AI research. We conducted a scoping review to map how Canadian researchers share patient-level data in multicenter medical AI collaborations. We searched PubMed, IEEE Xplore, ACM Digital Library, Scopus, and Web of Science from 2018 to February 2025 and implemented a human-in-the-loop large language model process to support screening and extraction, with reviewer validation. Among 3100 included studies, 160 reported multicenter patient-level data collection. Centralized pooling dominated this subset, with 95% of studies using centralized storage and 5% (n = 8) reporting decentralized approaches, including federated learning, sequential model transfer, and distributed feature sharing. Governance requirements were frequently described as multi-site and sequential, and 81.8% of multicenter collaborations reported parallel ethics approvals from three or more institutional review boards. Only one decentralized collaboration operated entirely within Canada. International partnerships comprised 80% of multicenter studies, and many cohorts included non-Canadian sites or non-Canadian data. Our findings support adoption of distributed model development protocols and interoperable governance that limit central pooling while enabling consistent training, validation, and reporting across sites, as only 1 of 160 multicenter studies reported a decentralized approach with Canadian patient data only.

1. Introduction

Precision public health relies on large-scale and diverse health data to target interventions and improve population outcomes and health equity [1]. However, sharing patient-level health data across institutions and jurisdictions remains a major challenge. Regulatory fragmentation, strict privacy laws, and incompatible technical systems impede the integration of data needed for collaborative analyses [2,3]. Without interoperable systems, AI models trained on isolated local datasets may fail to generalize to new populations. Such narrow training data can embed bias and lead to uneven performance among demographic groups [4]. Canada's multi-jurisdictional health system reveals these barriers. The country is a federation of ten provinces and three territories, each responsible for its own healthcare data. Provincial privacy laws and uneven digital infrastructure further fragment data governance. Therefore, the Canadian case is a strong example of cross-jurisdictional data sharing challenges [4]. Similar forms of decentralized health data governance exist in many countries. Therefore, lessons from the Canadian context can inform broader efforts to advance patient-level AI collaboration.

Collaboration in research enables investigators to integrate complementary expertise, datasets, and computing infrastructure [5–7]. Such collaboration expands interdisciplinary capacity to address complex questions that isolated teams cannot tackle [7,8]. Over the past decade, high-income countries have invested in multicenter consortia

* Corresponding author.
E-mail addresses: o.jafarinezhad@mail.utoronto.ca (O. Jafarinezhad), zahra.shakeri@utoronto.ca (Z. Shakeri).

https://doi.org/10.1016/j.artmed.2026.103422
Received 2 January 2026; Received in revised form 25 March 2026; Accepted 2 April 2026
Available online 21 April 2026

that share data and analytic tools, especially for precision-health initiatives [9]. Although Canada has officially endorsed this agenda in national strategies, recent reports suggest that Canadian laboratories exchange patient-level datasets less often than their international counterparts [9,10]. Geographic distance, provincial stewardship of health records, varying privacy legislation, and uneven digital infrastructure continue to create barriers to joint research between regions.

Modern medical science depends on sharing data and conducting multicenter studies, and top journals now mandate data availability to maximize the return on expensive clinical datasets [11]. However, in healthcare settings, it remains unusually difficult to transfer data between sites. Clinical records are sensitive and subject to legal, ethical, and technical restrictions that limit cross-institutional exchange. These safeguards protect patients and reduce harm, and the constraints tighten as data become more granular [12,13]. *Patient-level data* are observations on individual participants that carry a risk of re-identification [13]. Such data are used to train most medical AI models, including electronic health records (EHRs), radiological images (e.g., MRI and CT scans), biosignals (e.g., ECG waveforms), and genomic sequences [14]. Models trained on these large and detailed datasets can detect patterns that conventional statistics might miss, enabling precision care at the scale of a single patient [15]. Accordingly, patient-level data resources have catalyzed the rapid spread of artificial intelligence (AI) techniques in nearly every discipline of medicine [16]. Nevertheless, limited access to heterogeneous datasets may jeopardize equity, because algorithms developed on narrow or homogeneous samples often perform inconsistently for underrepresented populations.

Researchers have recently developed medical foundation models that leverage enormous datasets to perform multiple clinical tasks [17]. Some of these models even integrate multiple modalities of patient-level data (e.g., combining text reports with medical images) [18–20]. Although these models show promise across diverse healthcare applications, their development depends on collaboration among universities, hospitals, and research centers that can provide complementary datasets. Fragmented data governance in Canada limits such collective efforts and slows progress toward equitable AI innovations. In 2024, Canadian policymakers introduced Bill C-72 as a legislative framework to enable secure, interoperable medical data exchange and to prohibit data blocking by health information vendors [21]. However, the success of this legislation will likely depend on a clear understanding of existing collaboration patterns and on deploying privacy-preserving data sharing techniques at scale.

Several technical approaches have been proposed to support multicenter AI development without transferring raw patient-level data across institutional boundaries. Federated learning enables collaborative model training across distributed sites by exchanging model parameters or gradients rather than patient data [22]. In healthcare settings, federated learning has been extended to support semi-supervised representation learning, cross-institutional knowledge transfer, and robustness in adversarial environments, including emerging applications involving large language models [23,24]. Differential privacy provides formal mathematical privacy guarantees by introducing calibrated noise during training or prior to data or model release [25]. Secure multiparty computation and related cryptographic techniques enable joint computation (e.g., secure aggregation) while preserving the confidentiality of each site's raw inputs [26,27]. Together, these approaches constitute the principal technical paradigms for enabling privacy-preserving collaboration in multicenter medical AI and therefore motivate our focus on decentralized collaboration architectures in this review.

Although recognition is growing, no comprehensive assessment has studies how Canadian researchers share patient-level data or how often and with whom they collaborate across institutions, or to what extent they employ privacy-preserving approaches (e.g., federated learning) in these collaborations. Existing assessments lack scalable methods to examine thousands of records across jurisdictions under strict privacy constraints, and this limitation sustains the knowledge gap. To address this gap, we conducted a national-scale scoping review (2018-Feb 2025) using a custom human-in-the-loop large language model (LLM) pipeline. Our pipeline was necessary to handle the national scale and complexity of this review. In particular, the initial corpus included 245,886 articles with multilingual affiliations and heterogeneous data types across institutions, so a purely manual review was infeasible. Unlike generic approaches, our pipeline integrated retrieval-augmented generation (RAG) and multimodal models to extract information from figures and disambiguate text. Furthermore, we incorporated a dual-pass manual review and prompt validation to enhance the accuracy and trustworthiness of the pipeline outputs. Consequently, our pipeline distilled 245,886 records to 3100 studies meeting all inclusion criteria and yielded the first comprehensive national dataset on patient-level data sharing collaborations. Although our focus is Canada, barriers such as stringent privacy regimes, fragmented governance, and data heterogeneity are widespread in other federated health systems. Therefore, insights from this study remain relevant across federated health systems.

2. Material and methods

The overall implementation pipeline for this scoping review is illustrated in Fig. 1. To ensure scalability across a broad search space while reducing decision subjectivity, we developed and deployed a human-in-the-loop LLM pipeline. This review initially captured 245,886 PubMed records (2018–February 2025) focused on medical AI studies utilizing patient-level data. Given that a corpus of this magnitude is infeasible to review manually, our novel pipeline served as a scalable 'assistive layer' integrated into all major PRISMA-ScR stages, including search, screening, and data extraction. This pipeline integrated retrieval augmented generation (RAG) [28], LangChain integration, a multimodal model (i.e., LLaMA 7B combined with LLaVA for figure interpretation), and reviewer-guided prompt engineering. Incorporating these components into our review process supported reproducible processing at high volume and produced structured outputs for downstream synthesis. This approach aligns with a growing trend in biomedical research to use LLMs as assistive tools, and our review both adopts and advances that trend by fully integrating human-guided AI throughout the review process [29]. Fig. 1 illustrates this full LLM-enhanced pipeline, with each step color coded to distinguish tasks performed by human reviewers versus tasks driven by LLM.

2.1. Search strategy and selection criteria

We searched PubMed, IEEE Xplore, ACM Digital Library, Scopus, and Web of Science for this scoping review. For each search string, we screened at least the first 10 pages of Google Scholar and continued until no new records appeared. PubMed captured all records that we identified in the other bibliographic databases during screening. We developed the search terms from core concepts, then refined them through pilot searches and bigram analysis of 20 included articles. All 20 articles met inclusion criteria with high certainty, which helped identify additional terms and update the search strategy. The strategy combined keywords for artificial intelligence, machine learning, and healthcare, and the full query appears in Supplementary Note 1. We applied no restrictions on study design within these parameters at this stage. We searched for publications from January 2018 to February 2025 as illustrated in Figs. 1 and 2. This timeframe coincides with the start of fundamental and regulatory frameworks in health informatics. The 2018 Data Strategy Roadmap established a primary foundation for digital health and data interoperability within Canada [30]. This period also includes a substantial increase in publication volume as patient-level machine learning became a standard practice [31].

We included English language studies that reported on AI and health research and utilized patient data. These studies required a

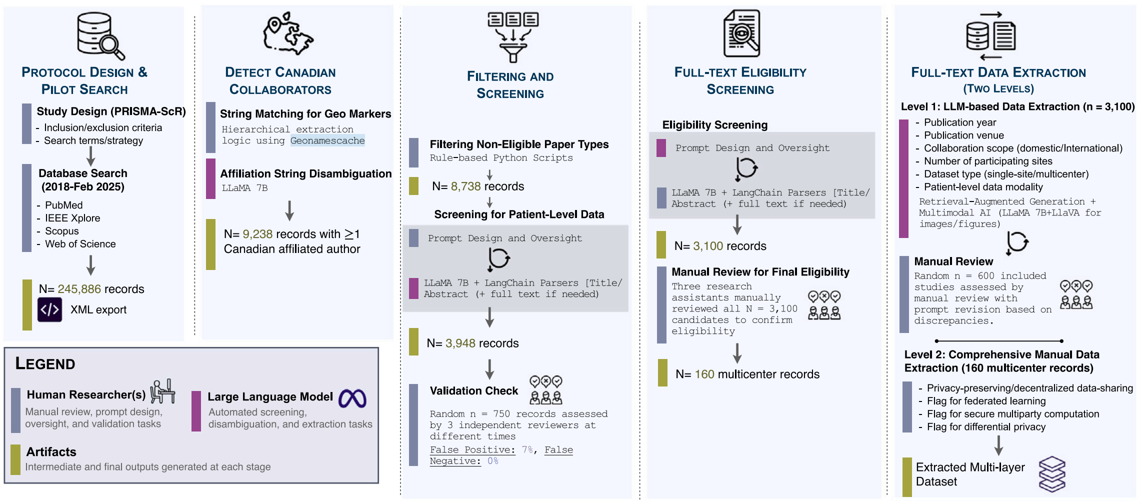


Fig. 1. Workflow of the human-in-the-loop screening and extraction process. This diagram illustrates the division of labor between human researchers and automated algorithms throughout the review process. The workflow progresses from the initial protocol design and database search to the final data extraction. Human researchers were responsible for prompt design, oversight, and the manual validation of intermediate outputs. Conversely, we deployed large language models to execute automated screening, disambiguation, and high-volume extraction tasks. The pipeline integrated LLaMA 7B with LangChain parsers to process textual data from titles and abstracts. Detailed manual reviews at distinct stages confirmed the accuracy of the automated eligibility and extraction logic.

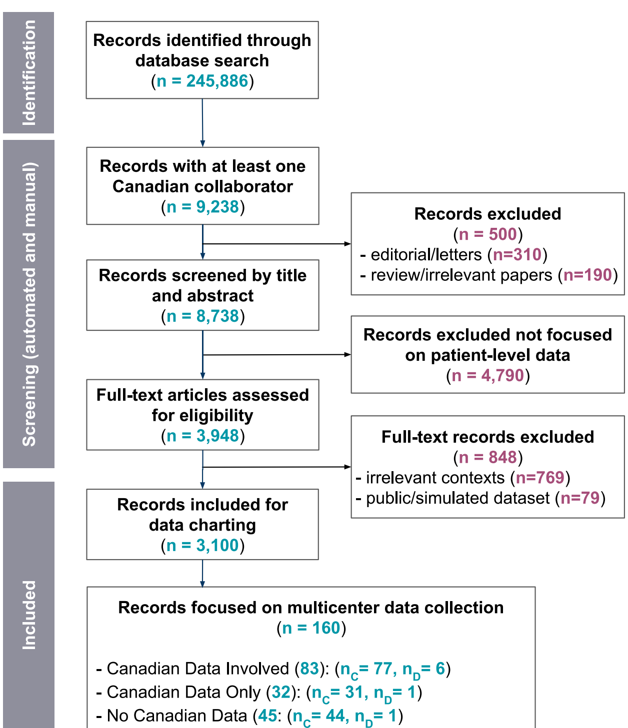


Fig. 2. The article selection process based on the PRISMA-ScR guideline. Among 115 studies with Canadian patient data, 83 included Canadian data alongside other countries (77 using centralized repositories (n_C), 6 using decentralized approaches (n_D)), and 32 studies used exclusively Canadian data (31 centralized, 1 decentralized). Among the 45 studies without Canadian data, 44 used centralized approaches and 1 used a decentralized approach.

Canadian affiliation to focus on the research context of multicenter collaboration within the domestic health system. We excluded reviews and opinion pieces and retracted publications. In this review, we define 'simulated data' as data created without underlying clinical patient records (e.g., from predefined mathematical models). The exclusion criteria were also applied to studies that used animal or simulated data. We excluded studies based solely on simulated datasets because they do not involve multicenter collection, coordination, or governance of real patient-level data, which is the focus of this review. We excluded studies that used synthetic data derived from real patient records unless they also involved multicenter governance of real patient-level data. We did not perform a formal evaluation of individual sources and included all records regardless of their quality. To manage the large volume of retrieved studies, we implemented a human in the loop LLM pipeline. We exported all retrieved records to XML format (Fig. 1) to allow a detailed and structural analysis of the collected data in subsequent stages.

2.2. Study selection process

All references retrieved from the search were exported for screening. We implemented a two-phase screening process to identify articles meeting our eligibility criteria. In the first phase, we filtered records by author affiliation to find studies with Canadian contributions. This was accomplished using a hierarchical workflow in which automated string matching for Canadian institution names was applied first, and an LLM pipeline (i.e., LLaMA 3 [32]) was used only to resolve cases that remained ambiguous after rule-based filtering. In such cases, LLM outputs were treated as supportive information, and final determinations were confirmed through manual review. Studies were classified as having Canadian participation if at least one author reported a Canadian institutional affiliation, including in cases of multiple affiliations. For multi-center studies, author affiliation and the geographic origin of the underlying data were extracted as separate attributes to distinguish Canadian institutional participation from the use of Canadian patient data (details of the affiliation detection algorithm are provided in Supplementary Note 1.1). In the second phase, we screened for the presence of patient-level data in each study. We applied an LLM-based pipeline (i.e., a 7-billion-parameter model integrated via LangChain [33]) to parse titles and abstracts, and full texts when necessary, for indications that the research used individual-level patient data. This pipeline flagged and retained studies explicitly involving patient-level clinical data and excluded those that only utilized fully de-identified, aggregate, or open-access datasets (e.g., studies exclusively based on public databases like MIMIC-III [34] were filtered out). To verify the accuracy of the automated first screening iteration, which

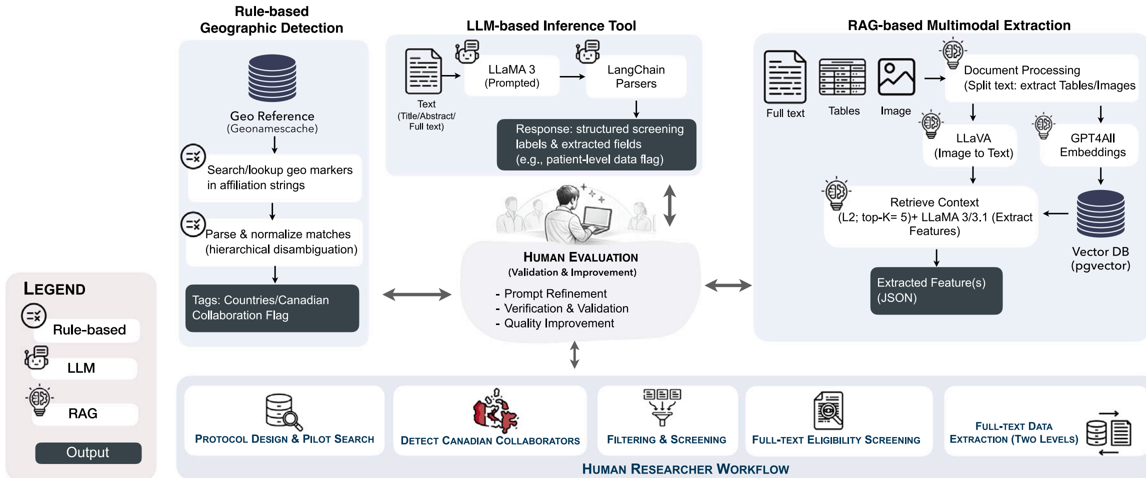


Fig. 3. Layered architecture of the human-in-the-loop assistive tool stack for screening and data extraction. The top layer depicts the assistive tool stack, including rule-based geographic detection, LLM-based inference, and a RAG-based multimodal extraction module. In the RAG pipeline, documents are chunked, embedded (GPT4AllEmbeddings), indexed in a vector database, and retrieved via similarity search. Human evaluation is centrally integrated, providing prompt refinement, validation, and quality control. The lower layer represents the research workflow. Assistive tools support these steps but do not replace final human decisions.

reduced 245,886 records to 3100, we conducted a validation check using a random subset of records. Three research assistants in our team with expertise in medical AI, computer science, and statistics involved in the inclusion and exclusion task manually assessed 750 records that were sampled from the full set of 245,886 retrieved studies. Sampling at this stage was intended to capture the heterogeneous mix of studies processed by the pipeline, including different publication years, clinical domains, and borderline or ambiguous cases. To evaluate whether the validation set covered the key decision scenarios, we verified that all four eligibility combinations were present in the sampled records (Canadian vs. non-Canadian affiliation ×patient-level data vs. no patient-level data).. We compared reviewer decisions with decisions from the LLM pipeline. The false positive inclusion rate was approximately 7%, and the analysis identified no false negatives. To directly quantify the risk of false negatives (i.e., eligible studies excluded before full-text review), we additionally audited a random sample of records that were excluded by the automated screening stage, including records near the model's uncertainty boundary. This audit complements the random validation of the full corpus and provides an estimate of residual false-negative risk among excluded records. We report sample-based confidence bounds and do not interpret zero observed false negatives as an absolute guarantee. Moreover, the absence of false negatives in the validation subset aligns with the conservative and recall-oriented configuration of our LLM-based screening pipeline, in which prompts and decision logic instructed the model to retain studies when eligibility was uncertain rather than exclude them. This result indicates that the recall-oriented configuration of the LLM settings was effective in minimizing premature exclusion of eligible studies during the initial screening stage. Records that met the automated inclusion criteria but did not meet reviewer criteria (e.g., false positives) were removed in later screening stages. The next screening phase used manual review of the 3100 included articles, which acted as a safeguard and reduced the likelihood that errors from the LLM-based screening stage affected the final study inclusion or results. The team further filtered the set to Canadian studies that used patient-level data in a multicenter setting. This process yielded 160 studies for detailed analysis. Among the 160 manually validated multicenter studies, eight (5%) implemented decentralized approaches. The broader set of 3100 studies remained part of the higher-level analysis because it contributed to the results and narrative of the review. Fig. 1 summarizes how the LLM supported this review and Fig. 3 describes the LLM algorithms and workflow used in our LLM pipeline.

2.3. Data analysis

We developed a structured data extraction form with two levels of abstraction. The form covered all 3100 articles and supported additional fields for the 160 multicenter studies. The team pilot-tested the form on 20 studies. The pilot informed revisions aligned with the study objectives and ensured the capture of all key data elements across the full set of included studies. Two reviewers manually extracted data from each included article. A customized LLM pipeline checked all extracted fields (Fig. 3). For all 3100 articles, we extracted bibliographic information such as publication year and venue. We also extracted study characteristics related to collaboration and data use. Collaboration scope was coded as *domestic* when all collaborating authors were affiliated with Canadian institutions. Collaboration scope was coded as *international* when at least one collaborating author was affiliated with an institution outside Canada (including dual affiliations on a single author). The number of participating sites was recorded for each study, along with whether the study used a single-site dataset or multicenter data collection. The patient-level data modality was recorded for each study. Categories included clinical imaging (e.g., MRI, CT, fMRI), electronic health records, biosignals (EEG, ECG), genomic or molecular data, and other reported modalities.

For the 160 multicenter studies, we recorded whether the dataset included Canadian patient data. We also recorded whether Canadian collaborators used only non-Canadian data sources. We also recorded any mention of privacy-preserving or decentralized data-sharing methods. Studies that reported federated learning, secure multiparty computation, or differential privacy were flagged. All data items were defined a priori, and these definitions were included in the LLM extraction prompts to support consistent capture across sources.

To reduce the risk of subjective decisions during data extraction, we developed a custom pipeline based on retrieval-augmented generation (RAG) [28] to support data charting. The pipeline used multiple LLMs, including `LLaMA 3`, `LLaMA 3.1`, and `LLaVA` [35]. We implemented the pipeline within the LangChain framework [33] to identify and record key study characteristics from the articles. The RAG pipeline followed a semi-structured and multi-vector retrieval strategy, in which documents were decomposed into smaller text segments. Embeddings were generated for each segment using `GPT4AllEmbeddings` within the LangChain framework. The resulting 384-dimensional vectors were indexed in a Pgvector database for similarity-based retrieval with L2 distance. Top-k retrieval was performed with k = 5, and the corresponding raw text or visual elements were retrieved and provided to the language models for extraction. This design increased the likelihood that extraction outputs were grounded in retrieved article passages rather than generated from embeddings alone. The pipeline also processed images extracted from the articles using `LLaVA`, which captured structured and visual information that automated reviews often miss. Before applying the pipeline to all included studies, we pilot tested and iteratively refined the extraction procedure on a subset of articles to ensure consistent capture of the target information. Before applying the pipeline to the full set of included studies (n = 3100), we pilot-tested and iteratively refined the extraction procedure on a random sample of 600 articles. Because many extracted variables were free text, chance-corrected agreement was calculated only for structured binary fields. Percent agreement and Cohen's kappa (κ) were calculated to quantify consistency between model outputs. Percent agreement represents the proportion of matching records but can be inflated in imbalanced data. Agreement was substantial for identifying use of patient-level data (96.5%, $\kappa = 0.62$), moderate for use of federated learning methods (79.7%, $\kappa = 0.43$), and fair for classification of centralized versus decentralized data repositories (73.9%, $\kappa = 0.38$). These values indicate higher consistency for well-defined variables and greater variability for classifications that required interpretation of study design, data location, and analysis workflow (e.g., classification of centralized versus decentralized). Accordingly, the RAG pipeline was used as an assistive tool, and final accuracy was determined through manual review. This manual validation step took place during full-text data extraction and evaluated the accuracy of LLM-assisted extraction. The team used discrepancies to refine prompts. We then compared LLM outputs for the 600 articles with the corresponding manual extraction results. The discrepancy analysis informed further refinements to improve consistency and generalizability of the extraction process. The feature extraction pipeline and tools are described in Supplementary Note 2.2.

To summarize data from the included articles, we used descriptive statistics and visual analytics. The cleaned dataset was analyzed to determine frequencies and proportions for key features (see Supplementary Note 2.3 for preprocessing details). These features included the percentage of studies with international collaboration and the distribution of data modalities across projects. We examined temporal trends to observe changes in collaboration patterns from 2018 to 2024. Moreover, we mapped the countries of partner institutions to visualize the geographic distribution of international collaborations. This analysis also ranked the most frequent collaborator countries outside Canada. We performed exploratory clustering and cross-tabulation analyses to explore relationships between study characteristics. For example, we grouped studies by data modality and clinical domain to identify common patterns.

We also performed a qualitative analysis of the full text of the 160 multicenter studies. We applied *open coding*, defined as the initial line-by-line labeling of relevant text segments with descriptive codes, to assign preliminary codes to the extracted material. This was followed by *axial coding*, which involves grouping and linking these initial codes into higher-level categories and identifying relationships among them to develop a coherent thematic structure [36,37]. Tables 1, 2, 3, and 4 summarize these results. This review identified common challenges and enablers of data sharing. These factors included requirements for multiple ethics board approvals, data standardization issues, and governance frameworks. The qualitative review complements the quantitative findings.

3. Results

We conducted a comprehensive literature search and screening process following the PRISMA-ScR guidelines [38], which we selected for its alignment with our review objectives over alternative frameworks [39–41]. The initial query yielded 245,886 records, and filtering for at least one Canadian-affiliated author reduced this to 9238. After removing 500 non-research items (310 editorials/letters and 190 unrelated reviews), 8738 records remained for title and abstract screening. We excluded 4790 studies that lacked patient-level data, leaving 3948 articles for full-text evaluation. An additional 848 were excluded upon full-text review (769 irrelevant topics and 79 using only public or simulated datasets). Ultimately, 3100 studies met all inclusion criteria and were charted in our scoping review (Fig. 2). To ensure validity, all included studies underwent full-text verification by our team to confirm their relevance and the accuracy of extracted data. Within this final corpus, 160 articles described multicenter patient-level data collection efforts, involving multiple institutions. Among these multicenter collaborations, 115 (71.9%) involved Canadian patient data (i.e., 83 in combination with international data and 32 using exclusively Canadian data), whereas 45 (28.1%) did not use any Canadian patient data.

3.1. Trends in patient-level data collaboration

To characterize the scale and geographical scope of collaborations, we categorized each study by its author affiliations as domestic (i.e., all authors at Canadian institutions) or international (i.e., including at least one author outside Canada). For example, a domestic oral cancer outcome study spanned two Toronto hospitals (i.e., the University Health Network and Sunnybrook Health Sciences Centre) [a1]. In contrast, the international ENIGMA consortium on obsessive-compulsive disorder pooled data from 18 sites across 16 countries [a2]. We found a strong overall trend toward international collaborations, nearly two-thirds of the included studies involved foreign collaborators. Specifically, out of 3100 projects, 1989 (64.2%) were international collaborations (with cross-border co-authors), whereas 1111 (35.8%) were purely Canada-only efforts (Figs. 4(a) and 4(b)). This pattern is consistent with national advisory reports that highlight persistent fragmentation in Canada's health data landscape impeding purely domestic research ties [42]. International partnerships clearly dominated the Canadian AI research output during our study period.

Annual collaboration activity grew substantially over time, with both domestic and international studies increasing in parallel through 2022. The total number of Canadian-affiliated multicenter studies rose from 195 in 2018 to a peak of 574 in 2022. Activity then dipped to 418 in 2023 before rebounding to 546 in 2024. International collaborations consistently outnumbered domestic ones each year, accounting for roughly 80% of all multicenter projects (Fig. 4(a)). Domestic collaborations peaked at 211 studies in 2022 and slightly declined thereafter (to 197 in 2024) despite targeted national programs (e.g. the CIHR Catalyst Grants) intended to stimulate intra-provincial data consortia. The 2023 downturn coincided with the end of that pilot funding and the redeployment of many research personnel to pandemic-related clinical backlogs. By 2024, a renewed federal investment in AI research corresponded with the observed publication resurgence. This temporal fluctuation aligns with broader trends in scientific output during and after the COVID-19 pandemic, which saw a transient contraction in research productivity [43]. Our inclusion of an automated LLM-based screening pipeline did not bias these temporal patterns, as a sensitivity analysis restricting to the confirmed 3100 studies reproduced the same yearly trajectory. Overall, international projects remained the majority throughout, peaking at 363 in 2022 and comprising the bulk of the post-2022 recovery. This emphasizes that Canadian-led precision health AI efforts have been largely global in reach.

Because our scoping review excluded studies using only public or simulated data, the collaborations we captured typically required sharing sensitive patient information governed by multiple jurisdictions. In practice, these projects faced substantial governance challenges, including strict data localization laws and complex cross-border patient consent processes. Our qualitative analysis of full-text methods

Table 1
Summary of multicenter AI collaborations with Canadian participation (2018–2024).

Domain	Modality mix	Collab scope-data mgmt(#)	Privacy tech	Explainability	Primary tasks	Interop/deploy
Cancer & oncology a1, a5, a6, a10, a11, a12, a13, a14, a15, a16, a17, a18, a19, a20, a21, a22, a23, a24, a25, a26, a27, a28, a29, a30, a31, a32, a33, a34, a35, a36, a37, a38	Clinical records a1, a12, a13, a16 a6, a19, a20, a23, a24, a25, a26, a27, a28, a29, a32, a33, a35, a37 Imaging data a5, a10, a11, a13, a14, a15, a16, a17, a18, a19, a20, a22 a6, a24, a25, a26, a27, a28, a29, a30, a31, a34, a36, a37, a38 Lab Data a16, a21, a24, a29, 31, a32 a10, a33, a34, a35 Survey data a12, a24 Genomic data a10	CA–C(2) a1, a12, a30, NoCA–C(pooled) a10, Intl–D(6) a11, Intl–D(71) a5, NoCA–C(11) a13, Intl–C(3) a14, a20, NoCA–C(3) a15, a21, Intl–C(15) a16, Intl–C(5) a17, a18, a19, a31, a35, NoCA–C(2) a22, Intl–C(pooled) a23, Intl–C(6) a24, Intl–C(2) a25, a27, a29, Intl–C(4) a6, a26, CA–C(4) a28, a36, NoCA–CA(6) a32, NoCA–C(5) a33, CA–C(5) a34, Intl–C(12) a37, NoCA–C(3) a38	Privacy-preserving distributed learning a11 Federated learning a5 REDCap for secure data capture a12,	Global interpretability[Gini importance a1; transparent coefficients a6, a20, a21, a32, a32, a33; RF importance a13, a14, a25, a29, Feature ranking a30], SHAP explanations[a12, a37], Imaging saliency [CAM a18, Heatmap localization (AIDA) a36]	Clf[a10, a11, a13, a14, a18,a19, a21, a29 a6, a31, a34, a36, a37, a38] Pred[a1, a11, a15, a16, a20, a22,a25, a26, a27, a28, a30 a6, a32, a33, a35] Seg[a5, a17] Risk/Stratify[a12, a16, a20, a23, a31, a32] Enablement/Infra[a24]	Shared code (GitHub) [a5, a6, a25] Web tools [a20]
Cardiovascular & vascular diseases a7, a39, a40, a41, a42, a43, a44, a45, a46, a47, a48, a49, a50, a51, a52, a53, a54, a55, a56, a57, a58, a59, a60, a61, a62, a63	Clinical records a39, a41, a44 a7, a46, a47, a48, a49, a50, a51, a54, a56, a57, a58, a59, a60, a61, a62, a63, Imaging data a40, a41, a42, a43, a45, a46, a47, a48 a50, a51, a53, a54, a55, a57, a59, a62, a63 Lab data a44, a49, a53, a61 Biosignal data a7, a52, a61, a62 Genomic data a60	NoCA–C(+700) a39, NoCA–C(4) a40, Intl–C(38) a41, Intl–C(9) a42, a47, Intl–C(2) a43, a62, CA–C(3) a44, Intl–C(12) a45, Intl–C(3) a46, a57, CA–C(2) a48, a59, NoCA–C(2) a49, a53, Intl–C(5) a50, a51, NoCA–C(15) a52, Intl–C(29) a54, Intl–C(10) a55, CA–C(9) a56, Intl–C(44) a58, CA–C(12) a60, Intl–C(4) a61, a63, CA–D(7)a7	Federated learning + differential privacy a7	SHAP explanations [a53, a56], Feature importance [XGBoost gain a39, a44, information gain a47, RF/GBM importance a49], Imaging saliency [Saliency mapping of ECG traces a62]	Clf[a7, a40, a42, a44, a45, a47, a52] Pred[a39, a41, a49, a55,a56, a57, a58, a59, a60, a61, a62] Seg[a43] Risk/Stratify[a46, a48, a50, a51 a53, a54, a56, a63]	Shared code (GitHub) [a7, a39, a60] Web tools [a58]
Neurological disorders a64, a65, a66, a67, a68, a69, a70, a71, a72, a73, a74, a75, a76, a77, a78, a79, a80, a81, a82, a83	Clinical records a66, a70, a72 a74, a79, a80, a83 Imaging data a65, a67, a68, a69, a70, a71, a73 a75, a76, a77, a78, a81, a82, a83 Lab data a79 Digital devices a64, a74 Biosignal a65, a67, a69 Genomic data a80 Survey data a66, a82	NoCA–C(4) a64, a75, Intl–C(25) a65, Intl–C(12) a66, Intl–C(3) a67, Intl–C(30) a68, Intl–C(608) a69, NoCA–C(3) a70, NoCA–C(pooled) a71, Intl–C(pooled) a72, NoCA–C(2) a73, a74, Intl–C(5) a76, Intl–C(2) a77, a82, CA–C(24) a78, CA–C(31) a79, Intl–C(4+) a80, Intl–D(83) a81, Intl–C(22) a83	Client-side encryption + server-side encrypted storage a64 Distributed learning a81	Global interpretability [a64, a66, a70, a73, a74, a83] SHAP explanations [a69, a79] Imaging saliency [Saliency maps (SmoothGrad) a71]	Clf[a64, a67, a70, a76, a78, a81] Pred[a66, a69, a71, a74, a77, a82, a83, a64] Seg[a68, a75] Risk/Stratify[a65, a72, a73, a74, a77, a79 a80, a82]	Shared code (GitHub) [a77, a81, a82]
Psychiatric & Mental Health a2, a3, a84, a85, a86, a87, a88, a89, a90, a91, a92, a93, a94, a95, a96, a97, a98, a99	Clinical records [a88, a89, a90, a92, a93, a94, a95, a96,a99] Imaging data [a2, a3, a86, a87, a88, a91, a92, a93, a94, a95] Survey data [a84, a85, a86, a90, a95 a97, a98, a99] Genomic data [a84, a89, a94] Lab data [a87] Environment data [a84]	NoCA–C(18) a2, Intl–C(13) a3, NoCA–C(21) a84, Intl–C(2) a85, Intl–C(8) a86, Intl–C(36) a87, CA–C(6) a88, NoCA–C(10) a89, NoCA–C(6) a90, Intl–C(30) a91, Intl–C(7) a92, Intl–C(21) a93, NoCA–C(3) a94, Intl–C(3) a95, a99, CA–C(17) a96, Intl–C(29) a97, Intl–C(40) a98		Global interpretability [a2, a3, a84, a85, a86, a87, a89, a90, a91, a93, a94, a96, a97, a99] Local interpretability [SHAP a92, K-LIME a2]	Clf[a2, a3, a87, a91] Pred[a85, a86, a88, a89, a90, a93, a94, a96, a99] Risk/Stratify[a84, a92, a95, a97, a98]	Shared code (GitHub) [a84, a96] ENIGMA's harmonized processing [a2, a3, a87]

(*continued on next page*)

descriptions revealed that 81.8% of the multicenter studies had to obtain parallel ethics approvals from at least three different institutional boards, adding significant administrative overhead and delay. For example, a 13-site ENIGMA neuroimaging study procured separate Research Ethics Board (REB) approvals at each institution before aggregating MRI data [a3]. Similarly, a German–Canadian kidney transplant study had to secure separate ethics approvals in Montréal, Vancouver, and Berlin to enable data exchange [a4]. Such requirements illustrate the bureaucratic burden inherent in multi-site health research.

In our multicenter subset, only 32 of the 160 collaborations (20%) relied exclusively on data from Canadian patients, while 128 (80%) included data from foreign sites. More than 40% of the international partnerships in this subset involved data from the United States (US) (43.8%), which was the most frequent non Canadian data source. This heavy reliance on foreign data sources reflects the need for Canadian researchers to tap into external datasets in the absence of a unified national data sharing platform. Encouragingly, it also aligns with international policy frameworks that promote cross-border health data interoperability (e.g., the USMCA digital trade provisions and the G7 2021 commitment to open health data standards [44,45]). However, without a national trusted-access network, Canadian teams often must depend on foreign repositories or multinational consortia to obtain the diverse patient data needed to train robust AI models. A recent OECD review noted that fragmented approval processes in one Canadian hospital stretched a cross-border data request to 18 months, until process re-engineering shortened it to under 3 months [46]. Such cases make clear the importance of translating these lessons into a cohesive Canadian governance strategy that can provide timely domestic access to richly varied patient data for research.

3.2. Data modalities in collaborative studies

We next reviewed which types of patient-level data were being shared and analyzed, as the choice of data modalities directly determines the achievable scope of precision public health applications. Fig. 5 quantifies the geographic distribution of collaborations and identifies frequent partner countries alongside exchanged data types. Our analysis confirms that imaging data constituted the most common modality in multicenter collaborations from 2018 to 2024. Structured clinical variables served universally as supporting covariates in these studies to facilitate robust multivariable modeling efforts. Radiological images, including MRI and CT scans, were frequently exchanged between institutions to support large-scale model training. Functional MRI specifically appeared in more than 22% of collaborations to drive neuroimaging research initiatives. For example, one national brain tumor consortium pooled MRI scans from dozens of distributed sites [a5].

Table 1 (*continued*).

Domain	Modality mix	Collab scope-data mgmt(#)	Privacy tech	Explainability	Primary tasks	Interop/deploy
Respiratory & lung conditions a100, a101, a102, a103, a104, a105, a106, a107, a108, a109, a110, a111, a112	Clinical records [a100, a101 a104, a105, a107, a109, a112] Imaging data [a102, a103, a106 a108, a111, a112] Survey data [a104] Genomic data [a100, a106] Lab data [a100, a107] Biosignal data [a110] Environment data [a104]	NoCA–C(8) a100, Ca–C(48) a101, CA–C(2) a102, a107, NoCA–C(2) a103, a112, Int-C(4) a104, NoCA–C(105) a105, NoCA–C(4) a106, Intl–C(3) a108, Intl–C(785) a109, CA–C(3) a110, Intl–C(5) a111		Global interpretability [XGB gain a101; LR coeffs, tree MDI, SVM permutation a104] Imaging saliency [Grad-CAM a102; CAM a103; Grad-CAM++a108]	Clf[a101, a102, a103, a104, a108, a110, a111] Pred[a107] Risk/Stratify[a100, a105, a106, a109, a112]	Shared code (GitHub) [a111]
Kidney & renal diseases a4, a113, a114, a115, a116, a117, a118	Clinical records [a4, a113, a114, a115, a116, a117, a118] Genomic data [a4] Lab data [a4, a113, a114, a115, a116, a117, a118]	Intl–C(4) a4, Intl–D(4) a116, Intl–C(3) a113, CA–C(2) a114, CA–C(201) a115, CA–C(pooled) a117, Intl–C(24) a118	Privacy-preserving FL and private blockchain (Ethereum DLT) a4	Global interpretability [a114]	Clf[a4] Pred[a4, a113, a114, a117] Risk/Stratify[a115, a116, a118]	Docker packaging and REST API a4
Infectious diseases a119, a120, a121, a122, a123, a124	Clinical records [a119 a121, a122, a123, a124] Imaging data [a119, a120] Lab data [a121, a122, a123, a124] Genomic data [a122]	Intl–D(20) a119, NoCA–C(5) a120, CA–C(7) a121, NoCA–C(4) a122, CA–C(19) a123, Intl–C(350+) a124	FL (FedAvg)a119	Imaging saliency [CAM a120]	Pred[a119, a120, a122] Risk/Stratify[a121, a123] Enablement/Infra[a124]	Shared code (GitHub) [a120] Released on NVIDIA NGC (Clara Train) a119
Critical care a125, a126, a127, a128, a129	Clinical records [a125, a126, a127, a128, a129] Digital devices [a128, a129] Lab data [a125, a126, a127, a128, a129]	NoCA–C(pooled) a125, Intl–C(20) a126, CA–C(pooled) a127, NoCA–C(4) a128, CA–C(16) a129		Global interpretability [a126, a128]	Clf [a125, a127, a129] Pred [a126, a128, a129]	Shared code (GitHub) [a129]
Surgical & trauma conditions a8, a130, a131, a132, a133	Clinical records [a131, a132, a133] Imaging data [a131] Genomic data [a8, a130, a132]	Intl–C(10) a8, Intl–C(7) a130, NoCA–C(20) a131, Intl–C(8) a132, NoCA–C(3) a133		Global interpretability [a8, a131], SHAP explanations [a133]		Web page calculator a133
Reproductive, pregnancy, and neonatal health a134, a135, a136, a137, a138	Clinical records [a134, a135, a136, a137, a138] Lab data [a135, a136, a138]	NoCA–C(10) a134, Intl–C(3) a135, Intl–C(17) a136, Intl–C(49) a137, Intl–C(2) a138		Global interpretability [a134, a135, a138], SHAP explanations [a137]	Clf[a137] Pred[a134, a135, a138, a136]	
Gastroenterology and hepatology a139, a140, a141	Clinical records [a139, a140] Lab data [a139, a140] Imaging data [a141]	NoCA–C(4) a139, Intl–C(23) a140, Intl–D(2) a141	FL a141	Global interpretability [a139]	Pred[a141] Risk/Stratify[a139, a140]	Web calculator [a140]
Other a9, a142, a143, a144, a145, a146, a147, a148, a149, a150, a151, a152, a153, a154, a155, a156, a157, a158, a159, a160	Clinical records [a9, a142, a143, a144, a145, a146 a149, a151, a152, a153, a154, a156, a157, a158, a159] Lab data[a143, a9 a151, a153, a156, a158, a159] Imaging data [a9, a143, a146, a148, a149, a160] Survey data [a143, a146, a147, a150, a153, a155] Genomic data [a9] Digital devices [a150]	Intl–C(2) a142, CA–C(16) a143, CA–C(pooled) a144, CA–C(3) a145, Intl–C(2200) a9, NoCA(4) a146, Intl–C(69) a147, NoCA–C(2) a148, Intl–C(pooled) a149, NoCA–C(21) a150, CA–C(5) a151, NoCA–C(74) a152, CA–C(2) a153, NoCA–C(17) a154, NoCA–C(15) a155, Intl–C(3) a156, Intl–C(10) a157, NoCA–C(7700) a158, Intl–C(>2) a159, NoCA–D(8) a160	Risk-based anonymization a9 OMOP CDM v5 across sites a157 Distributed learning a160	Global interpretability [a143, a149, a151, a155, a156, a158, a145, a146, a159], Imaging saliency [CAM a148]	Clf[a145, a148, a159] Pred[a142, a143, a147, a149, a150, a160] Risk/Stratify[a151, a158] Enablement/Infra[a9, a144, a146, a152, a153, a154, a155, a156, a157]	Deployed private cloud (OpenStack)a9 Shared code (GitHub) [a156, a159, a160]

Abbreviations: **CA** Canada-only; **Intl** international (including Canada); **NoCA** No Canadian data; **C** centralized; **D** decentralized; **FL** federated learning; **Clf** Classification; **Pred** Prediction; **Seg** Segmentation.

Similarly, a Canada–US lung cancer screening collaboration shared low-dose chest CT images to develop radiomics models [a6]. Major partners including the United States and the United Kingdom frequently exchanged these imaging datasets (Fig. 5). The prevalence of medical imaging likely reflects the rich diagnostic utility of these high-dimensional data sources. Furthermore, the global adoption of standardized formats such as DICOM significantly simplifies technical integration between diverse institutions. This dominance also stems from the structural compatibility of image data with advanced deep learning architectures. Methods such as Grad-CAM enable the visualization of image-based predictions to directly enhance the interpretability of algorithmic outputs [47].

Although medical images are widely used, their application introduces significant technical and logistical challenges for researchers. The generation of ground-truth annotations for imaging data remains a labor-intensive bottleneck for clinical experts. Furthermore, stringent privacy regulations effectively restrict the cross-border sharing of high-fidelity scans between international institutions. These regulatory and technical factors significantly complicate the construction of large multi-institutional datasets for robust validation [48]. Many hospital archiving systems store images in proprietary formats that require substantial preprocessing efforts prior to analysis. Consequently, these raw data necessitate extensive curation before researchers can utilize them for model development. Other data modalities appeared less frequently in our review but hold substantial promise for precision medicine applications. Specifically, genomics and biosignals (e.g., EEG) offer unique opportunities for developing targeted therapeutic interventions and diagnostics. Genomic data enable personalized diagnostics to identify specific molecular markers that drive disease progression. Conversely, biosignals support the real-time monitoring of physiological states to detect acute changes in patient condition. However, the integration of these complex sources presents distinct computational and analytical challenges for current algorithms. High-dimensional health data modalities, including genomics and other biosignals, often involve many features relative to available samples, a setting commonly described as the "curse of dimensionality," which may introduce statistical and validation challenges [49–51]. In distributed environments, particularly in multicenter contexts with site-level heterogeneity, these challenges may further affect model stability [a7]. In addition to algorithmic challenges, multicenter genomics and biosignal collaborations often require substantial harmonization of measurement protocols, preprocessing pipelines, and consent/governance processes across sites,

Table 2
Primary task catalogue by disease domain in multicenter AI collaborations with Canadian participation (2018–2024).

Domain	Primary tasks
Cancer & oncology a1, a5, a6, a10, a11, a12, a13, a14, a15, a16, a17, a18, a19, a20, a21, a22, a23, a24, a25, a26, a27, a28, a29, a30, a31, a32, a33, a34, a35, a36, a37, a38	**Clf** [mesothelioma subtype a10; overall survival (OS) + HPV status a11; primary cancer type (BM patterns) a13; molecular subgroup a14; tumor detection + pediatric subtype a18; OS prognosis + ICI short/long survivors a19; MCED detection + site a21; radiogenomic subtype (BRAF vs others) a29; HPV triage a31; benign vs malignant nodules a6; anastomotic leakage risk a33; Cancer detection (micro-US, biopsy targeting) a34, multicancer subtype across sites a36, tumor subtype a37; Skin cancer detection (malignant vs benign) and multiclass lesion classification a38] **Pred** [LOS after oral cancer surgery a1; survival a11; CVD risk from calcifications a15; OS/DFS + surgery benefit a16; OS + local/regional control a20; local failure post-SBRT a22; BM progression after SRS a25; 2y OS + lifetime risk a26; PD-L1 + PFS stratification a27; PFS biomarker under ICI a28; PFS + OS (1st-line) a30; LNM/recurrence after resection a32; nodule malignancy a6; csPCa detection + biopsy guidance a35] **Seg** [glioblastoma sub-compartment boundaries a5; PET tumor volume a17] **Risk/Stratify** [traumatic stress prevalence/severity + predictors a12; benefit-based groups a16; risk triage a20; outcomes + care-quality (STEMI cancer) a23; surgery vs surveillance stratification a32; HPV reproducibility emphasis a31] **Enablement/Infra** [LHS analytics + dashboards + ML readiness a24]
Cardiovascular & vascular diseases a7, a39, a40, a41, a42, a43, a44, a45, a46, a47, a48, a49, a50, a51, a52, a53, a54, a55, a56, a57, a58, a59, a60, a61, a62, a63	**Clf**[obstructive CAD a40, a42; HF case ID a44; Systolic/diastolic dysfunction a45; CAD rule-out a47; LVEDP class a52; Multi-disease ECG a7] **Pred**[30-day MALE/death a39; MACE w/ plaque+FFRCT a41; readmission AMI a49; Echo LV/RV quant a55; operative strategy (arch repair) a56; CAD triage (CAC+CCTA) a57; Warfarin dosing a58; AAA rupture a59; bleeding risk a60; OMI/culprit vessel a61; LV/RV dysfunction a62] **Seg**[MI segmentation (cine MRI) a43] **Risk/Stratify**[MACE hybrid MPI a46; AAA features ESUS a48; REFINE SPECT missing-data robustness a50; MACE risk (sex diffs) a51; AAD mortality a53; Plaque nomograms a54; Aortic arch repair (death/stroke) a56; Mortality (body composition CTAC) a63]
Neurological disorders a64, a65, a66, a67, a68, a69, a70, a71, a72, a73, a74, a75, a76, a77, a78, a79, a80, a81, a82, a83	**Clf**[PD vs controls a64; VNS responder vs non-responder a67; Benign vs malignant PNST a70; ALS vs controls a76; ADHD vs control; ASD vs control a78; PD vs controls (T1 MRI) a81] **Pred**[MCID at 6/12/24 mo a66; Fast vs slow RT a69; Brain age (regression) a71; Progression/clinical change a74; Functional outcomes from network markers a77; Age/sex prediction a82; Chronic motor function a83; Motor-impairment severity proxy a64] **Seg**[Brain extraction/skull-strip (FLAIR) a68; 10-class tissue segmentation (DWI) a75] **Risk/Stratify**[Structure–function links a65; Phenotyping/cohort stratification a72; Lesion–symptom mapping; sex-specific a73; Trial stratification/enrichment a74; Disease subtyping/progression a77; Cohort discrimination; risk signatures a79; Genetic association & fine-mapping a80; Normative deviation associations a82]
Psychiatric & mental health a2, a3, a84, a85, a86, a87, a88, a89, a90, a91, a92, a93, a94, a95, a96, a97, a98, a99	**Clf**[OCD vs controls a2; Bipolar vs controls a3; OCD vs HC; subgroup classifiers a87; MDD vs controls a91] **Pred**[Alcohol use a85; Depression onset (2- & 5-yr) a86; Brain-age gap → ADT response a88; Lithium response a89; Differential remission across 5 monotherapies and 3 combinations a90; Future psychosis (CHR) a93; rTMS response (≥20% PANSS-NS reduction) a94; Binary remission + best drug recommendation a96; Predict non-adherence risk (≥ 12 sessions definition) to target peer-support a99] **Risk/Stratify**[cognitive ability mediates effects of genes & environment on PLEs a84; Phenotyping/staging (case–control + illness duration) a92; Brain-age deviation ↔ cognition in MDD a95; Relationship quality a97; HCP mental health risk actors during COVID-19 a98]
Respiratory & lung conditions a100, a101, a102, a103, a104, a105, a106, a107, a108, a109, a110, a111, a112	**Clf**[Identify COPD a101; Normal vs abnormal LUS a102; Patient sex from CXR a103; physician-diagnosed asthma a104; Sliding-artifact detection a108; Wheeze vs normal a110; Multiclass disease classification a111] **Pred**[COPD admission & 30-day readmission a107] **Risk/Stratify**[Genetic variant ↔ IgE a100; Resource use & expenditures a105; Subtype discovery a106; Protein intake ↔ MV outcomes/weaning a109; Mortality risk stratification in ILD using CT-based ML score a112]
Kidney & renal diseases a4, a113, a114, a115, a116, a117, a118	**Clf**[Donor–recipient molecular compatibility a4] **Pred**[Short- & long-term graft failure, mortality, rejection a4; 1–5 yr kidney failure & all-cause mortality a113; CKD progression (40% eGFR decline or kidney failure, 1–5 yr) a114; 30-day adverse outcomes a117] **Risk/Stratify**[Identify undertreated CKD pts a115; TG patient stratification & risk of graft loss a116; Prognosis for ≥40% eGFR decline or kidney failure (1–3 yr) a118]
Infectious diseases a119, a120, a121, a122, a123, a124	**Pred**[Oxygen requirement, MV/death a119; CXR severity/trajectory a120; Reaction occurrence a122] **Risk/Stratify**[Immune response post-booster a121; Comparative effectiveness of CAP regimens a123] **Enablement/Infra**[Validate case-finding algorithms, screening strategy a124]
Critical care a125, a126, a127, a128, a129	**Clf**[Prolonged MV >14d a125; Delirium from notes a127; 30d mortality/AKI classification under scarce data a129] **Pred**[ICU mortality a126; Early ARDS mortality (≤24h) a128; ICU/hospital LOS regression a129]
Surgical & trauma conditions a8, a130, a131, a132, a133	**Clf** [T-cell–mediated rejection a8; Molecular diagnosis of ABMR/TCMR (±injury)a132; ABMR/TCMR diagnosis, ensemble report a132], **Pred** [graft loss a8; post-op transfusion need (THA)a133], **Risk/Stratify** [phenotype discovery of rejection/injury vs histology/DSA a130; frailty via CT muscle score for 30-day outcomes a131]
Reproductive, pregnancy, and neonatal health a134, a135, a136, a137, a138	**Clf**[vaginal birth after dinoprostone IoL a137] **Pred**[severe neonatal morbidity (<32 wks) a134, postnatal gestational age estimation a135, semen concentration upgrade (post-varicocelectomy) a138, AI-driven prognostic counseling (IUI/IVF conversion) a136]
Gastroenterology and hepatology a139, a140, a141	**Clf**[detect hepatic steatosis S0 vs. ≥S1 a141] **Risk/Stratify** [ED-arrival triage for in-admission adverse outcomes a139; 30/90-day mortality risk aiding steroid/LT triage decisions a140]
Other a9, a142, a143, a144, a145, a146, a147, a148, a149, a150, a151, a152, a153, a154, a155, a156, a157, a158, a159, a160	**Clf**[High vs low acuity (CTAS 1–3 vs 4–5) for remote triage a145; detect elbow/shoulder dislocation on X-rays a148; distinguish GCA (active/inactive) vs controls from plasma a159] **Pred**[Multi-task clinical prediction with few-shot label-efficiency a142; predict disease course/remission risk; subtype joint-involvement patterns a143; regress contact-avoidance, hygiene-maintenance, and policy-support scores a147; predict recurrence after Bankart repair a149; predict likelihood of wearable use in children a150; retinal-age regression; centralized vs FL vs TM a160] **Risk/Stratify** temporal-cluster detection to stratify deterioration risk a151; non-response risk and burden (HRU changes) a158] **Enablement/Infra** [Re-identification leakage from shared embeddings; anonymization comparison a144; Research-ready data platform to capture/anonymize/harmonize multi-source clinical data a9; Best practices for handling missingness in supervised ML a146; ICU spillover effects; counterfactual capacity-expansion simulations a152; Implement AE & lipid indicators, identify practice gaps, and support counseling a153; Real-world initiation of second-line antihyperglycemics by region/CVD risk a154; Measure prevalence/incidence at population scale a155; Compare diagnosis- vs lab-based labels; quantify cross-site variability a156; Second-line antihyperglycaemics—3/4-point MACE outcomes a157]

Abbreviations: **Clf** Classification; **Pred** Prediction; **Seg** Segmentation.

Table 3
Frequently used AI techniques (> 10 Studies) in multicenter Canadian studies with a central data repository (n = 160).

AI Technique(s)	#(%)
Neural networks	36 (23.7%)
Random forest	35 (23.0%)
Logistic regression	33 (21.7%)
Gradient boosting	30 (19.7%)
Support vector machines	30 (19.7%)
LASSO/Ridge/Elastic net	20 (13.2%)
Survival models	19 (12.5%)
Principal Component Analysis (PCA)	15 (9.9%)
Decision trees/CART	10 (6.6%)

which can raise the operational threshold for collaboration. In our full-text coding of multicenter studies, harmonization of protocols and preprocessing pipelines was more frequently described for imaging collaborations than for genomics or biosignals, which may contribute to the observed modality imbalance. We interpret this as a likely contributor rather than a definitive explanation. We anticipate that future studies will incorporate these underused modalities as multimodal AI approaches continue to mature.

3.3. Multicenter data collection collaborations

Our analysis of 160 multicenter studies reveals a distinct preference for international partnerships over domestic networks. Specifically, 128 collaborations (80%) included at least one international site, whereas 32 (20%) remained exclusively Canadian. Fig. 6 illustrates the temporal divergence between these domestic and international collaboration categories. International activity accelerated markedly after 2020 and culminated in 43 distinct collaborations during 2024. This surge likely reflects a shift toward cross-border partnerships to address global health challenges. Conversely, domestic activity peaked in 2023 before declining slightly in the subsequent year. Detailed domain and task summaries are provided in Tables 1 and 2.

Table 4
Comparison of multicenter decentralized data repositories in Canadian healthcare research.

	Bogowicz 2020 a11	Dayan 2021 a119	Schapranow 2023 a4	Bakas 2022 a5	Souza 2024 a81	Agrawal 2024 a7	Qi 2024 a141	Nielsen 2024 a160
Clinical domain/task	Head & Neck cancer: HPV & 2-yr OS	COVID-19: O_2 need (24/72h)	Kidney transplant CPMs (risk)	Glioblastoma: tumor boundaries (ET/TC/WT)	Parkinson's disease classification	ECG multi-label diagnosis	Hepatic steatosis detection	Retinal age prediction (RAG)
Participating sites (scope)	6 cohorts (multi-institution)	20 institutions (global)	Germany + Canada (major centres)	71 sites across 6 continents	83 centres (global; many small)	7 hospitals (province-wide)	2 centres (private + public; simulated clients)	UK Biobank (up to 2400 simulated clients) + BRSET
Population/sample size	1174 patients	~16,000 ED presentations	~8,000 retrospective cases	6314 cases	1817 scans	1,565,849 ECGs/243,128 patients	n = 208 (153 + 55); 2080 images	16,630 (UKB) + 958 (BRSET) images
Modality	CT radiomics	CXR + EMR (vitals, labs)	Clinical; HLA typing + matching; labs; pathology	mpMRI	3D brain imaging	12-lead ECG time-series	Ultrasound (B-mode) + biopsy steatosis grades	Fundus photos via RETFound features
Distributed model	Distributed feature selection + logistic regression	Federated learning (FedAvg) “EXAM”	Federated learning on private blockchain	Federated consensus segmentation	Traveling Model (TM); sequential site updates	FL with Differential Privacy (DP-SGD)	Simulated FL (four algorithms incl. FedAvg)	Foundation-model-driven deep learning; compare FL vs TM
Infrastructure/deployment	EuroCAT/DistriM-style distributed setup	NVIDIA FL framework; multi-site federation	Private blockchain; consortium pipelines	Global aggregation server; international consortium	Sequential TM; no central aggregation	Multi-hospital secure servers; site-wise eval	Flower-based simulation; partition strategies	8-bit features for efficiency; low-resource friendly
Reported performance (high-level)	Distributed ≈ centralized (no AUC drop; identical LR coefficients)	AUC > 0.92; improved generalizability vs single-site	N/A (protocol)	DSC ↑ vs public model — local: +27% (ET), +33% (TC), +16% (WT); OOS: +15% (ET), +27% (TC), +16% (WT)	TM AUROC ≈ 83% vs 80% centralized	FL (81.03%) comparable to pooled (82.07%); benefits for small-data sites	FedAvg AUC 0.93; central 0.90; single-site 0.83; imbalance hurts	MAE ~3.6 years; TM converges faster; FL/TM ≈ centralized
Explainability methods	Transparent LR coefficients	-	-	-	-	-	-	–
External/out-of-sample validation	Internal validation across cohorts; no dedicated held-out institutions	Independent external hospitals (incl. largest test site); multi-site evaluation	-	6 held-out sites (OOS) for generalization	Real-centre TM; compared to centralized baseline; no separate external set	Internal across 7 hospitals; no external institutions	Held-out test split; no external centres (simulated clients)	External BRSET evaluation in addition to UK Biobank
Client heterogeneity & non-IID handling	Multi-cohort radiomics; no explicit non-IID method	Inter-site EMR/CXR heterogeneity; assessed via multi-site training & external testing (no special non-IID algorithm)	Cross-national heterogeneity anticipated; harmonization/QC pipelines specified (protocol)	Scanner/site variability addressed by standardized mpMRI preprocessing; OOS sites held out	Many small, heterogeneous real centres; TM chosen for non-IID/minimal-data settings	Hospital-wise label prevalence differs; FL competitive with pooled, esp. for low-data sites	Explicit non-IID experiments (class imbalance & partitioning) quantify performance drop	Simulated client heterogeneity; TM converges with fewer local updates than FL

Abbreviations: **ED** = emergency department; **EMR** = electronic medical record; **CXR** = chest X-ray; **mpMRI** = multi-parametric MRI; **ET** = enhancing tumor; **TC** = tumor core; **WT** = whole tumor; **OOS** = out-of-sample (held-out sites); **TM** = traveling model; **FL** = federated learning; **DP-SGD** = differentially private SGD; **LR** = logistic regression; **AUC** = area under the ROC curve; **MAE** = mean absolute error; **CI** = confidence interval; **UKB** = UK Biobank; **BRSET** = Brazilian Multilabel Ophthalmological Dataset.

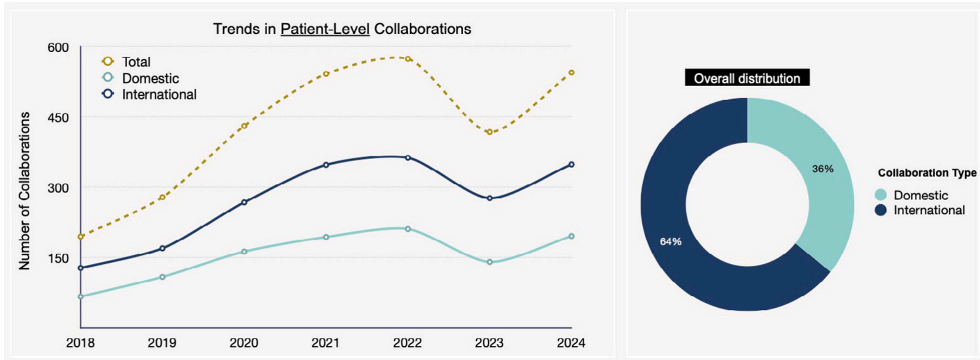
Consistent with the broader dataset, in which imaging appeared in over 22% of studies, more than half of multicenter collaborations incorporated imaging data (82, 51.2%), including MRI in 30 (18.7%) and CT in 25 (15.6%). In most cases, imaging was used alongside other data rather than alone (31, 19.4% image-only vs 53, 33.1% multimodal), which shows its role as a key analytical modality frequently paired with clinical and laboratory information. Clinical data appeared in 102 (63.8%), but only 19 (11.9%) studies were clinical-only and 83 (51.9%) were multimodal. Genomic data appeared in 14 (8.7%) studies. This distribution reflects the dominance of imaging methods in collaborative research and indicates a growing, although still limited, use of genomic data in multi-site studies, consistent with the broader shift toward precision medicine and genomics-driven care. For example, a multinational multicenter lung-transplant study involving 10 centers in six countries used microarray-based gene expression profiling and machine learning to derive biopsy-based molecular T-cell-mediated rejection (TCMR) scores, with higher scores associated with subsequent loss of graft [a8]. Likewise, a pharma-academic alliance assembled a centralized and research ready database that integrates clinical records, MRI, and omics across large multicenter trial cohorts (e.g., 35,000 multiple sclerosis (MS) patients and >230,000 MRIs), which allows imaging omics analyses at scale [a9].

We profiled data modalities and disease relationships using co-occurrence analysis and hierarchical clustering. We standardized terminology across studies to group related data types. For example, terms such as 'genomic sequencing' and 'gene expression' were combined under the label 'Genomic Data'. Patient-reported outcomes and quality-of-life measures were grouped as 'Survey Data'. We then constructed a binary matrix that captured the presence of each modality across disease categories. We applied hierarchical clustering to the resulting co-occurrence matrix to identify patterns of modality use (Fig. 7). The clustering results suggested several modality specific profiles by disease area. Cancer and Oncology (56) spanned multiple modalities but relied most on imaging (25), followed by clinical (18) and laboratory data (10). Cardiovascular and Vascular Diseases (45) combined clinical (19) and imaging (17) data, with distinct use of biosignals (4). Psychiatric and Mental Health (32) relied heavily on survey data (8), alongside imaging (10) and clinical data (9), and was one of only two areas that used environmental data. Neurological Disorders (30) showed the broadest mix, with activity in seven modalities.

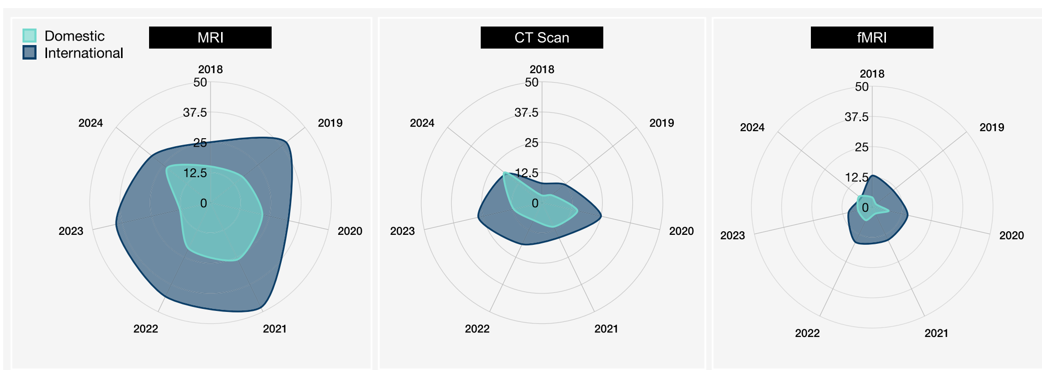
Genomic and molecular data appeared less frequently but were present in select disease areas. These modalities contributed to Surgical and Trauma Conditions and Psychiatric and Mental Health studies, where genetic information can support stratification and treatment selection. Laboratory biomarkers were also prominent in Cancer and Oncology and Kidney and Renal Diseases, where test results guide diagnosis and monitoring. Imaging and EHR data dominated across disease domains, with a growing use of genomic and laboratory data in areas that benefit from molecular signals.

3.4. Centralized data repositories in multicenter collaborations

A key question for multicenter AI projects is how patient data are shared and aggregated across institutions. We therefore investigated the data sharing architectures used in the 160 collaborative studies to



(a) Annual counts of domestic and international (Canada-non-Canada) collaborations using patient-level data with overall proportions.



(b) Domestic and international collaborations by imaging modality (MRI, CT, fMRI).

Fig. 4. Collaborations involving Canadian institutions that use patient-level data. Domestic collaborations involve only Canadian partners, whereas international collaborations include at least one partner outside Canada. (a) shows yearly collaboration counts from 2018 to 2024 together with their overall proportions, and (b) presents the distribution of collaborations across three imaging modalities (MRI, CT, and fMRI) for both domestic and international categories. These visualizations use full calendar year groupings for 2018–2024 to ensure consistency and comparability.

distinguish between centralized versus decentralized repository models. In a centralized model, patient-level data from multiple sites are pooled into a single, consolidated database or repository that authorized investigators can query. We found that this centralized approach was by far the dominant architecture, with 152 out of 160 multicenter collaborations (95%) in our review used a central data repository to gather data from all participating sites. Only a small minority of studies (8 collaborations, 5%) pursued decentralized data sharing. Many centralized projects reported following best practices to facilitate cross-site integration. For example, adherence to the Findable, Accessible, Interoperable, and Reusable (FAIR) principles for data sharing was commonly emphasized to harmonize formats and definitions across institutions [52]. At the same time, concentrating sensitive patient information in one location raises well-known privacy and security concerns, which the studies addressed through measures such as data de-identification, encryption during transfer, and strict access controls for the central repository. As shown in Fig. 2, of the 152 centrally coordinated studies, 31 were conducted using exclusively Canadian patient data, whereas the remainder included data from at least one international site. Nearly all centralized collaborations described obtaining multiple institutional ethics approvals and related agreements. Most projects required clearance from the REBs of each participating organization, given that data were being pooled centrally. Several retrospective studies noted that they received informed consent waivers because only de-identified or previously collected clinical data were used to minimize risk to patients. Moreover, some teams developed de-identification pipelines that removed or masked personal identifiers before aggregation to ensure compliance with privacy regulations (e.g., Ontario Personal Health Information Protection Act) and streamline data ingestion [53].

3.4.1. Dataset scope and ethics/REB practices

We observed that the scale of datasets pooled in centralized Canadian collaborations varied widely, which reflects both the opportunities and limitations of multi-institutional data gathering. Among the 32 studies that relied exclusively on Canadian patient data, roughly one-third (34.4%) had fewer than 1000 patient records in their analysis. These were often pilot projects or studies of rare conditions where available data were inherently limited. An additional 31.3% of studies used medium-sized cohorts of 1000–10,000 patients, typical of multi-center research on common conditions (e.g., multi-hospital studies in breast cancer or heart failure). A smaller share of projects managed to assemble large datasets. Four studies (12.5%) each included on the order of 10,000–100,000 patients, and another four (12.5%) analyzed between 100,000 and 1,000,000 patient records. Three ambitious nationwide initiatives (9.3%) studied cohorts exceeding one million individual records, demonstrating that 'big data' efforts are feasible in Canada, although rare. This wide variation in dataset scope indicates barriers to accessing very large integrated datasets, but also shows that some collaborative networks have achieved population-level scale. In fact, only a few projects leveraged truly massive cohorts, whereas many Canadian multicenter studies operated with relatively modest sample sizes. Consequently, the persistence of smaller datasets, despite interest in big data, aligns with ongoing efforts to develop machine learning methods that perform well with limited data [54,55]. Indeed, several included studies explicitly focused on data-efficient algorithms or data sharing techniques to compensate for small sample sizes.

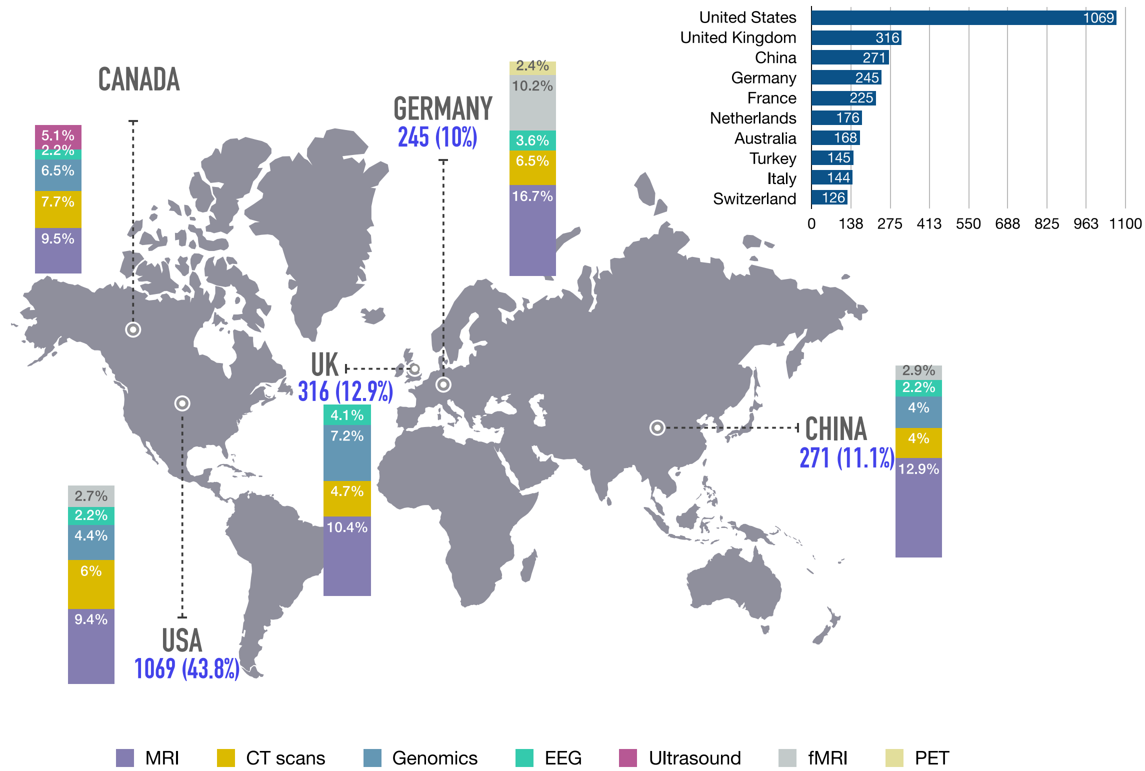


Fig. 5. International Collaborations in Medical AI Involving Canada from 2018 to 2024 (full calendar years). Geographic distribution of patient-level data collaborations between Canada and international partners. The bar chart (top-right) ranks the ten most frequent partner countries, while stacked bars beside selected nations show the five most common data modalities exchanged with Canada (e.g., Canada-all partners, Canada–US).

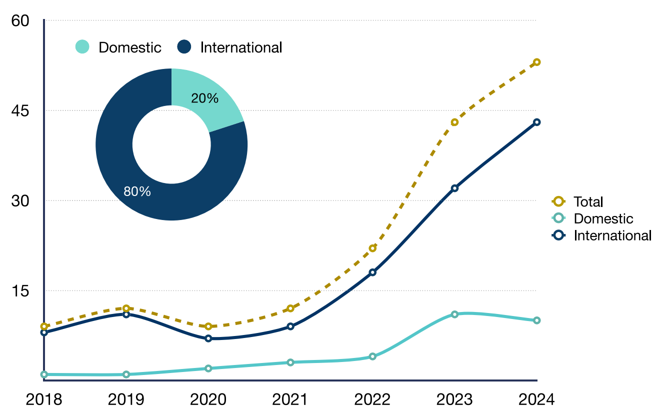


Fig. 6. Trends in Domestic and International Multicenter Collaborations, from 2018 to 2024 (full calendar years, n=160). Annual counts of Canadian-led multicenter collaborations are shown for domestic, international, and combined totals. The inset doughnut summarizes the full period: 80% international and 20% domestic. Overall activity rises sharply after 2022, driven mainly by international partnerships, while domestic collaborations peak in 2023 before declining slightly.

3.4.2. AI methods

Multicenter studies used several AI algorithms that ranged from regression models to deep learning architectures. Neural networks were the most common model family and appeared in 36 studies (23.7%), followed by random forests in 35 studies (23%) (Table 3). These two approaches were frequently chosen for their ability to handle high dimensional data. Neural nets supported complex modalities such as medical images or text, and random forests showed robust performance on heterogeneous tabular clinical data. Logistic regression were almost equally prevalent (33 studies, 21.7%), consistent with its continued

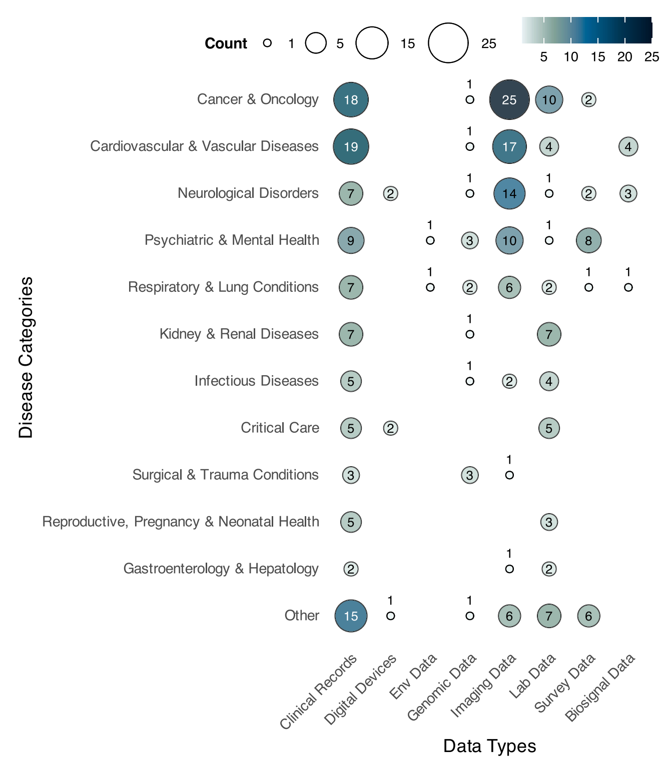


Fig. 7. Bubble plot of disease categories by data modalities in Canadian multicenter data collection collaborations (160 studies). Bubble size and color indicate the number of studies, with larger and darker circles representing higher counts. Cancer & oncology and cardiovascular diseases dominate collaborations, with imaging and clinical records being the most frequently used data types.

role as a baseline in clinical prediction due to its interpretability and well-calibrated risk estimates. In fact, logistic regression has long been a standard in clinical decision support [56,57], and evidence suggests that many specialized ML models do not significantly outperform logistic models in discrimination or calibration for health outcomes [57].

Ensemble methods such as gradient boosting (30, 19.7%) and support vector machines (30, 19.7%) were also widely applied, particularly for structured datasets where predictive accuracy was prioritized. Regularized regression techniques (e.g., LASSO, Ridge, Elastic Net; 20, 13.2%) were frequently chosen to manage high-dimensional clinical variables, while survival models (19, 12.5%) addressed the longitudinal and time-to-event nature of many health outcomes.

Dimensionality reduction methods like PCA (15, 9.9%) supported exploratory analyses in high-dimensional omics or imaging settings. Decision trees (10, 6.6%), although less common, provided simple and interpretable rules in clinical scenarios that required transparent decision making. These patterns show that researchers tend to choose methods based on the problem at hand and align the model complexity with data type and clinical context rather than defaulting to the newest techniques. In practice, this often meant prioritizing reliable and interpretable algorithms that clinicians could trust over more experimental approaches pursued for novelty alone.

Only two projects used cutting-edge foundation models, indicating that such large-scale pre-trained AI systems have not yet been widely adopted in Canadian collaborative health research. One study applied a foundation model for structured EHRs pretrained on 2.57 million patient records and reported that local adaptation matched gradient boosting performance while being 60%–90% more sample-efficient [a142]. In the imaging domain, another study employed a histopathology foundation model with adversarial Fourier-based domain adaptation (AIDA) across four cancer types, which enhanced cross-site generalization compared to standard normalization approaches and generated interpretable heatmaps that corresponded to pathologist annotations [a36]. These examples show the promise of foundation models to improve learning efficiency and generalizability. However, they were outliers in our review, as most collaborations relied on more conventional modeling techniques.

3.4.3. AI explainability, and deployment

We also reviewed the extent to which studies reported using explainable AI (XAI) techniques to interpret model predictions. Despite broad recognition of the importance of transparency in clinical AI, less than half of the collaborations explicitly documented any XAI approach. Only 63 of the 152 centralized-data studies (41.4%) mentioned incorporating methods to explain or interpret the model outputs. Among those who did, the majority focused on global interpretability tools that provide overall feature importance or model-level insights. Global feature importance rankings (e.g., regression coefficients or feature importance scores from tree-based models) were the most common explanation technique, used in 44 studies (28.9%). In contrast, local and instance-level explanations were rare, with only 10 studies (6.5%) applied post-hoc methods like SHAP or LIME to generate patient-specific explanations for predictions. Similarly, visual saliency mapping for imaging models was reported in just 9 studies (5.9%). These typically involved heatmap overlays on medical images (using techniques such as CAM or Grad-CAM [47]) to highlight regions influential in the model's decision, sometimes validated by experts (e.g., radiologists or pathologists confirming that highlighted image features corresponded to known disease markers [a36]). A handful of other studies performed related analyses such as dimensionality reduction (e.g., PCA, t-SNE, UMAP) to inspect data clustering, or calibration and robustness checks (e.g., reporting Brier scores or feature stability under perturbation), which aid interpretability indirectly but are not dedicated XAI methods. Our findings indicate that global explanation methods (i.e., feature importance at the cohort level) predominated, especially in studies using tabular clinical data, whereas saliency maps were the preferred approach for imaging models.

Most studies remained at the proof of concept stage in real world deployment and data sharing infrastructure, and few delivered tools or systems for clinical use. As summarized in Table 1, the most common translational artifacts included open source code repositories and simple web based calculators. In 11 studies (7.2%), authors released code or algorithms publicly (e.g., a multicenter venous thromboembolism risk model on GitHub [a60]). In 4 studies (2.6%), authors provided web interfaces for risk calculators or decision support (e.g., an AI driven 90 day mortality risk calculator for alcoholic hepatitis as an online tool [a140]). Several collaborations deployed trained models in Docker containers and exposed RESTful APIs to support integration [a4, a24]. One infectious disease collaboration distributed pre trained models through the NVIDIA GPU Cloud platform with Clara Train [a119], which indicated an initial move toward cloud based model sharing in healthcare. Only a minority of studies reported formal interoperability standards for multi site data integration and reuse. Four collaborations adopted the OMOP Common Data Model (CDM) across participating institutions [a49, a142, a156, a157]. This approach supports harmonized data semantics and scalable analytics. These projects used OMOP or related standards to improve transfer and replication across settings. Most collaborations in this review prioritized model development and evaluation over clinical deployment. Teams often provided shared resources such as code, tools, and containers, while integration into workflows or electronic record systems remained limited.

3.5. Decentralized data repository in multicenter collaborations

Finally, we examined the uptake of privacy-preserving decentralized data sharing techniques, given their potential to overcome privacy and jurisdictional barriers. In a decentralized (i.e., federated) model, data remain at source institutions, and only computed updates or summaries are exchanged, which avoids centralizing raw patient data. Such approaches can improve scalability and maintain patient privacy in highly regulated environments like healthcare [58]. However, as of early 2025, the use of decentralized architectures among Canadian multicenter studies was still in its infancy. Only 8 of the 160 multicenter collaborations (5%) implemented any form of decentralized or federated learning approach, which indicates that these methods remain rare in current practice. We identified three distinct patterns among the decentralized initiatives. The most common was federated learning (FL), employed in 6 studies [a4, a5, a7, a119, a141, a160]. In a typical FL setup, each participating site trains a shared machine learning model locally on its own data and periodically sends model updates (e.g., gradient averages or weight deltas) to a central coordinator. The coordinator updates the global model and redistributes it to the participating sites. This strategy keeps data at each site and can achieve accuracy comparable to pooled data training. However, it requires coordinated training rounds, sufficient network bandwidth for repeated model exchange, and careful handling of non identically distributed data (e.g., class imbalance or scanner specific effects across hospitals). A second decentralized design, used in 2 studies, was the so-called 'traveling model' approach (also known as sequential learning) [a81, a160]. Instead of using a central aggregator, a single model instance is sent serially from one site to the next and updated at each site. This approach reduces coordination overhead and bandwidth demands because it does not require simultaneous communication with a server. However, the training order can affect the resulting model, and updates to the global model state tend to occur more slowly than in FL. The third pattern was a distributed classical analysis where one study [a11] kept data siloed at each site but shared derived features or summary statistics to build a combined model. This approach keeps the methodology simple and inherently interpretable (e.g., sites might all compute the same regression model on their own data and then aggregate the learned coefficients), but it comes at the cost of lower model capacity and necessitates extremely well-harmonized data preprocessing across sites.

Although adoption of decentralized methods was limited, these early studies showed that privacy-preserving collaboration is feasible and may offer practical advantages. Several reported that federated or distributed models achieved performance close to a pooled data benchmark, which indicates that multi site learning can proceed without direct data sharing (Table 4). In particular, the federated learning projects showed that training on a broader range of patient populations, distributed across hospitals or countries, improved the generalizability of AI models. This benefit was most evident for rare conditions, where no single center has enough cases to train a robust model. For example, one federated study on Parkinson's disease drew on data from 7 hospitals within one Canadian province to build a more diverse classifier [a81], and another combined multi-continent data to develop a global model for retinal disease prediction [a160].

At the same time, our analysis identified gaps and challenges in these decentralized efforts. None of the FL-based studies described a clear plan for long-term network sustainability. Most papers did not explain how a federated model would be maintained and updated after the study ended. Model explainability was also absent. No federated learning study reported using explainable AI methods to interpret model decisions, even though local interpretation is feasible. Only one collaboration was conducted entirely within Canada using a decentralized protocol (a cancer study using federated learning across Canadian sites [a7]). All other instances of federated or distributed learning involved international partners or data. This suggests that purely domestic federated networks in Canadian healthcare are still essentially nonexistent.

Even without a national federated network, Canadian researchers have begun to participate in larger international decentralized collaborations. Several teams in Canada have contributed to multinational FL initiatives that support secure analyses at scale, often in rare disease studies or international registries. These efforts can benefit Canadian research despite local data sharing hurdles. For example, one global federation used FL to analyze COVID-19 outcomes across dozens of hospitals (including Canadian sites) without moving any raw patient data [a119]. This type of project shows that cross border federated learning can work while keeping Canadian data in place and sharing only learned model parameters. In these implementations, each institution retains control over its sensitive data, and training proceeds through the exchange of model parameters or gradients that are aggregated centrally. This approach supports learning across sites without transferring identifiable patient information outside the source institution.

Nonetheless, interoperability issues stemming from siloed provincial health information systems continue to limit broader adoption of federated networks within Canada [59]. Our findings show a clear contrast between strong AI research capacity and constrained data sharing practices in Canada. Scalable and privacy preserving infrastructure for health data collaboration is required for the next phase of equitable and precision driven public health. Our results highlight recurring architectural and governance-related patterns associated with cross-institutional collaboration, including fragmented governance, multiple REB requirements, data localization laws, and limited use of federated methods. These observations are consistent with recent analyses of digital health innovation adoption, which suggest that implementation is shaped by interacting domains such as technological complexity, organizational capacity, regulatory context, and system-level sustainability [60].

4. Discussion

Our review was motivated by the need for broad data sharing in precision public health. Canada has made major investments in AI research, but its health data continue to be siloed among institutions and jurisdictions. Our scoping review reveals a clear contrast between Canada's advancing AI work and its limited data sharing practices. AI models are now more powerful than ever, capable of learning from text, images, genomics, and beyond, with documented gains in precision medicine. However, Canadian researchers face real hurdles in accessing and managing the diverse patient data needed to put this potential to work. This study found that despite available tools such as multimodal models and federated learning, Canadian teams rarely adopt these approaches in practice. Most collaborations still centralize data or analyze datasets held outside Canada.

4.1. Impact of patient-level data on precision medicine

In the studies we reviewed, patient-level data were foundational for enabling individualized care and deeper insights. In the multicenter collaborations involving Canada, imaging modalities were the most frequently used data type, mainly in cancer and cardiovascular applications. However, we observed an increasing inclusion of genomic data and patient-reported outcomes in recent projects. This trend toward multimodal datasets suggests that precision medicine research is expanding beyond purely biological signals to also encompass behavioral and social determinants of health. Such broader data integration represents a move toward a more comprehensive, whole-person approach to personalized care. For example, the United Kingdom's 100,000 Genomes Project successfully embedded whole-genome sequencing into routine care by linking each participant's genomic data with longitudinal clinical records in a secure, privacy-protected environment. This illustrates the power of integrating diverse patient-level data at national scale to accelerate discovery [61–63]. Similarly, the United States' All of Us program has already enrolled over 500,000 participants and links EHRs with genetic profiles to enable nationwide precision medicine studies [64]. Open resources like the UK Biobank, a cohort of 500,000 individuals with harmonized health data, have enabled discoveries worldwide by providing a large and standardized dataset [65].

In this context, our review finds that Canada's research ecosystem, while highly international, struggles to build comparable domestic data resources. Imaging data were often the easiest to share because radiology standards such as DICOM support multi site exchange. Many Canadian led AI projects have used imaging data. By contrast, genomics and biosignals remain underused in Canadian collaborations. Integration is challenging because genomic data require complex analysis and high computing capacity, and EEG and ECG signals vary between sites and include noise. Our review identified more collaborations that included genomics by 2024, which suggests a shift to multimodal work as data management tools improve. Canada still lags in this area. Without large unified national datasets, investigators often use non Canadian sources to obtain the heterogeneity needed for robust models. This pattern aligned with our finding that nearly two thirds of collaborations with Canadian involvement included international partners. Only a minority of the multicenter studies were conducted entirely within Canada. Interprovincial barriers, including misaligned privacy rules and technical incompatibilities, continue to limit large scale data pooling within the country.

4.2. International collaboration vs. Domestic fragmentation

A key finding of this review is that Canadian-led AI collaborations are mostly international. Partnering globally can facilitate knowledge exchange and give research teams access to larger or more diverse cohorts abroad, but it also exposes the limited capacity for purely domestic data sharing. In practice, Canadian scientists often rely on foreign data to achieve the sample size or diversity their models require. As a result, if most algorithms are trained on non-Canadian populations, they may not generalize well to Canadian patients. For example, a diabetic retinopathy model trained largely on images from abroad could underperform on Canadian patients if demographic or environmental factors differ. We noted several studies that could have been done entirely within Canada if data access had been straightforward, but they relied on international partners to reach the required scale. Ironically, it is sometimes simpler for a Toronto researcher to collaborate with a Boston team under a common framework than to navigate separate agreements with colleagues in Montreal or Vancouver. In other words, it can be easier to collaborate across an ocean than across provincial lines.

These findings indicate an urgent need to strengthen intra-Canadian research linkages. Robust national data resources and networks could support AI models and precision health insights tailored to Canada's population and health system. It would also help ensure that Canadian patients directly benefit from research conducted in Canada. Some steps in this direction are already underway. Legislative efforts such as Bill C-72 aim to promote secure, interoperable health data exchange across provinces, while prohibiting vendor practices that block data sharing. Pan-Canadian initiatives are beginning to reduce fragmentation by aligning governance and infrastructure. For example, the Marathon of Hope Cancer Centres Network is linking major cancer centers to share clinical and genomic information, with the goal of creating a 15,000-patient 'gold cohort' for precision oncology. Success in these initiatives will depend on technology and policy, as well as institutional trust and sustained collaboration. International partnerships will remain important, but stronger domestic data sharing is the path for Canada to move from mainly supporting others' advances to leading transformative work in health AI.

4.3. The potential of LLMs to enhance healthcare research

We also see the rise of large language models, especially new multimodal models as a promising development for collaborative healthcare research. As clinical datasets grow to include diverse data types (e.g., text, images, signals and physiological signals), advanced LLMs can integrate these heterogeneous inputs and reveal complex patterns that single-modality models might miss. Recent work illustrates this potential. Google's Med-PaLM 2 model achieved expert-level performance on US medical licensing exam questions, which supports high-quality clinical question answering by an AI [66]. GatorTron was trained on over 90 billion words from de-identified electronic health records and achieved strong accuracy for medical concept extraction and relation identification [67]. In pathology, PathAlign used more than 350,000 pathology images paired with text and produced diagnostic reports judged acceptable in about 78% of cases, with key clinical details retained and lower manual workload [68]. In radiology, an experimental system called RaDialog generated chest X-ray reports and answered clinician questions through an LLM-based interface [69]. These examples show how multimodal AI can support expert workflows and improve efficiency and consistency in radiology and pathology.

As LLMs advance, their ethical and safety implications require careful attention. Patient privacy is paramount. Large models can unintentionally memorize and regurgitate identifiable details from their training data, so safeguards are needed when models are trained on sensitive health information. Techniques such as differential privacy and careful filtering of training data can help mitigate this risk. Accuracy and accountability are also critical. Models like Med-PaLM 2 and RaDialog are impressive, but they are still prone to errors or nonsensical outputs and may miss important clinical context. At present, LLMs should serve as assistive tools for clinicians rather than autonomous decision makers. Human oversight, rigorous validation, and regulatory frameworks will be needed before these models can be fully trusted in frontline care.

4.4. Privacy-preserving collaboration

The case studies of federated learning in our review (Table 4) show that privacy-preserving techniques can enable collaboration that would otherwise be blocked by data sharing restrictions. For example, Bogowicz et al. trained a head-and-neck cancer outcome model between institutions in different countries without pooling any raw data, by periodically aggregating model updates from each site [a10]. Dayan et al. similarly developed a model to predict COVID-19 severity across 20 hospitals on several continents. They achieved high accuracy (i.e. AUC 0.92 for 24-h mortality risk), while each hospital retained full control of its own data [a119]. These studies indicate that FL can match the performance of traditional centralized models even under strict local privacy rules.

Within the multicenter subset (n = 160), formal differential-privacy (DP) mechanisms were rarely reported (*one* study mentioning DP; *one* implementing DP in a decentralized protocol). Among studies using decentralized data repositories in multicenter collaborations, we did not observe explicit use of secure multiparty computation, only one explicitly incorporated formal DP mechanisms (see Table 4). This limited uptake likely related to both methodological and socio-technical considerations. From a technical perspective, techniques such as DP-SGD introduce explicit privacy–utility trade-offs and require careful calibration of hyperparameters (e.g., clipping norms and privacy budgets), which may affect model convergence and predictive performance. Multi-hospital implementations combining FL and DP have reported observable performance trade-offs relative to non-DP federated [a63]. Beyond these technical challenges, the integration of DP adds operational complexity and coordination overhead across heterogeneous clinical sites. As articulated by Alami et al. drawing on the NASSS framework, adoption of advanced digital health innovations is influenced not only by technical feasibility but also by organizational readiness, governance alignment, regulatory context, and long-term sustainability considerations [60].

Traditional multicenter research relied on de-identified data warehouses. Platforms such as i2b2 and Canada's GEMINI network showed that hospitals could contribute de-identified patient records to a central repository for analysis across many sites [70,71]. De-identification reduces risk but does not guarantee privacy. For example, Sweeney re-identified a public figure's health record by linking an 'anonymous' medical dataset with voter records using only birth date and postal code [72]. This well-known incident revealed limits of simple de-identification and prompted development of privacy-preserving methods. Modern approaches include differential privacy, which adds statistical noise to query results to protect individuals while preserving aggregate patterns [25]. Other approaches include synthetic data generation, which produces artificial records that match key properties of the source data without exposing actual patient information [73]. Additional approaches include cryptographic methods, including homomorphic encryption and secure multiparty computation, which support analysis without exposing raw inputs.[26,27].

The federated learning collaborations we reviewed applied several of these techniques in practice. For instance, Schapranow et al. augmented a two-site kidney transplant FL project with a blockchain ledger to audit all data exchanges [a4], and Pati et al. incorporated a rare-case data augmentation strategy to ensure that even small hospitals' data influenced a global brain tumor model [a4]. All projects encrypted inter-site communications, and some sites normalized or stratified data locally to reduce variability between sites. Privacy-preserving collaboration is feasible, but it requires careful planning and sustained technical support.

4.5. Maintaining performance amid data heterogeneity and class imbalance

In the included studies, federated approaches generally achieved performance comparable to traditional centralized training. For example, Qi et al. reported that their federated liver ultrasound model obtained an AUC of 0.85–0.93 on different test sets, closely matching the performance of a model trained on pooled data [74]. Pati et al. likewise observed significantly improved accuracy in detecting glioblastoma tumor boundaries when training on a federation of 71 sites, compared to using a much smaller public dataset [a5]. Reported proximity between federated and centralized performance should be interpreted in light of each study's experimental design rather than as general evidence of performance equivalence. Detailed reporting on preprocessing harmonization, cohort matching, and distributional balance was not consistently available, and potential sources of selection, reporting, or implementation bias cannot be fully ruled out. At the same time, multi-institutional evaluations such as the DeCaPH framework [75] have documented performance degradation of approximately 3%–4% compared with centralized collaborative training in multiple clinical tasks, which supports the plausibility of this pattern. However, maintaining high model performance across disparate datasets is challenging. A key issue is data heterogeneity. If one hospital's imaging protocols or patient demographics differ greatly from another's, a single global model may not fit all data distributions well. In Qi et al.'s study, for instance, the model's performance dropped (i.e., AUC fell as low as 0.34) when one local site's data distribution diverged markedly from the global distribution [74]. This pattern highlights the need for data harmonization and, in some settings, site level model adaptation. It also suggests a practical advantage of centralized datasets. When data are pooled in one location, they can be merged and cleaned into a more uniform dataset before modeling. In federated settings, these inconsistencies require algorithmic handling during training.

Class imbalance is another common challenge. In many medical datasets, certain conditions or outcomes are relatively rare, which can skew training and reduce a model's generalizability. In a multicenter scenario, one hospital might contribute most of the rare cases while another contributes mostly controls. Without correction, the global model may become biased toward the majority class and underperform for minority classes. We saw this issue in one rare-cancer federation, where combining heterogeneous datasets without proportionate class representation led to high variability in the model's predictions [a4]. Researchers addressed these problems using minority oversampling and cost-sensitive learning to maintain fairness and sensitivity for rare outcomes in federated models. These approaches can reduce disparity, but they require careful validation in distributed settings.

4.6. Future directions for collaborative AI in healthcare

Our review found that 95% of multi-center studies used central data pooling while only 5% explored federated learning, indicating that privacy-preserving approaches remain underused. Concerted efforts are needed to scale federated learning across institutions, including common standards for data exchange and model training. One solution is to adopt shared frameworks (e.g., NVIDIA FLARE [76]), which provide adaptable pipelines while enforcing core processes across projects [76]. Inconsistent data schemas impede integration, but adopting common data elements and harmonized ontologies could alleviate this barrier [52]. Federated algorithms must also handle heterogeneous, non-IID data across sites to ensure models remain generalizable. Moreover, rapid progress in multimodal AI and large language models presents new opportunities. However, developing next-generation foundation models that integrate text, imaging, and other modalities will require data at scales beyond any single institution [17–20,77]. Collaborative approaches (e.g., federated training or secure model sharing) could enable this without exchanging raw data.

Beyond these technical issues, collaborative AI faces structural barriers. Data governance in Canada remains fragmented across provinces, which complicates multi-center data sharing. National initiatives aim to address this fragmentation, as a Pan-Canadian strategy calls for a unified health data system [42] and Bill C-72 aims to mandate interoperability [78]. However, many smaller or under-resourced institutions still lack the infrastructure and personnel needed to implement AI [79,80]. This disparity, if left unaddressed, could widen the divide between well-resourced centers and others and exacerbate inequities. Investing in shared infrastructure, workforce training, and no-code tools can help democratize AI participation and prevent these consequences. Finally, model transparency and real-world validation remain critical, as many advanced models operate as 'black boxes' that undermine clinician trust. Incorporating explainability methods and involving end-users can improve transparency [79]. Few such systems have been tested in live clinical workflows, so demonstrating their safety and effectiveness in real-world practice is an essential next step [81].

4.7. Limitations

This study has several limitations. First, we used a large language model to assist with the initial screening of nearly 245,000 references, which introduced a risk of classification errors. We mitigated this by having human reviewers manually check the model's decisions at different time points, as well as all $n = 3100$ included articles. The first evaluation process resulted in 7% false positives and no false negatives, where studies initially flagged by the LLM were excluded upon manual review as false positives. Second, inconsistent terminology required normalization. Authors used different terms for the same concept (e.g., CT and computed tomography, EEG and electroencephalography). We mapped synonyms to a shared vocabulary and grouped related items. This step improved consistency across the dataset, but it may have reduced specificity in some categories. Our clustering and categorization results are only as good as the input data coding. If our grouping overlooked subtle distinctions, then the patterns we reported could be affected. Third, we relied on a generative AI model (i.e., LLaMA 7B) to help extract features from study abstracts at scale. LLM outputs can be inconsistent or sensitive to formatting, so it is possible that ambiguous or poorly structured abstracts led to some misclassified or missed details. However, all LLM-assisted steps in our study were overseen by human researchers. We manually checked each data point that the AI extracted and reported only findings that were verified by our team with relevant expertise.

Despite these limitations, the major trends and observations identified in this scoping review are robust. Our work provides a useful overview of collaborative medical AI research in the Canadian context and highlights structural barriers that must be addressed. Many other federated healthcare systems face similar data governance challenges, so these insights apply beyond Canada. Without concerted intervention, Canadian researchers may continue to remain data contributors to others' studies rather than leaders of homegrown precision health innovations, potentially leaving Canadian patients underserved by advances made elsewhere. In other words, without improvements in privacy-preserving collaboration, certain Canadian populations may not benefit equally from AI-driven health innovations. This gap is an emerging public health equity issue.

CRediT authorship contribution statement

Omid Jafarinezhad: Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qian Zhang:** Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Ryan Rezai:** Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Mohammad Noaeen:** Writing – review & editing, Methodology. **Aviv Shachak:** Writing – review & editing. **Behrouz Far:** Writing – review & editing. **Zahra Shakeri:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Funding

This research was funded by the University of Toronto Data Sciences Institute (DSI-PDFY3R1P31), the Canadian Institutes of Health Research (CIHR) Artificial Intelligence for Public Health (AI4PH) Trainee Award, and the Natural Sciences and Engineering Research Council of Canada (NSERC) through the Canada Research Chairs Program and the Discovery Grant (RGPIN-2025-07037).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.artmed.2026.103422.

Data availability

The data supporting the findings of this study, along with code, and related materials are publicly available on GitHub at https://github.com/HIVE-UofT/Collaborative_Research_Review.

References

[1] Khoury MJ, Iademarco MF, Riley WT. Precision public health for the era of precision medicine. Am J Prev Med 2015;50(3):398.
[2] El Arab RA, Abdulaziz O, Sagbakken M. Economic, ethical, and regulatory dimensions of artificial intelligence in healthcare: an integrative review. Front Public Health 2025;13:1617138.
[3] Wang EH, Myles P, Foraker R, Toh S, Mosquera L, El Emam K, Burcu M. Crossing borders securely: synthetic data and federated networks for privacy-preserving access to real-world data and emerging use cases. Npj Digit Med 2025;8(1):758.
[4] Bubela T, Genuis SK, Janjua NZ, Krajden M, Mittmann N, Podolak K, Svenson LW. Medical information commons to support learning healthcare systems: examples from Canada. J Law, Med Ethics 2019;47(1):97–105.
[5] Wuchty S, Jones BF, Uzzi B. The increasing dominance of teams in production of knowledge. Science 2007;316(5827):1036–9.
[6] Venturini S, Sikdar S, Rinaldi F, Tudisco F, Fortunato S. Collaboration and topic switches in science. Sci Rep 2024;14(1).
[7] Lee S, Bozeman B. The impact of research collaboration on scientific productivity. Soc Stud Sci 2005;35(5):673–702.
[8] Qin J, Lancaster FW, Allen B. Types and levels of collaboration in interdisciplinary research in the sciences. J Am Soc Inf Sci 1997;48(10):893–916.
[9] of Health Research CI. CIHR strategic plan 2021–2031: Investing in the long-term health of Canadians. 2023, https://cihr-irsc.gc.ca/e/52331.html. [Accessed 30 May 2025].
[10] Kulasegaram K, Grierson L, Barber C, Chahine S, Chou FC, Cleland J, Ellis R, Holmboe ES, Pusic M, Schumacher D, et al. Data sharing and big data in health professions education: Ottawa consensus statement and recommendations for scholarship. Med Teach 2024;46(4):471–85.
[11] Flanagin A, Curfman G, Bibbins-Domingo K. Data sharing and the growth of medical knowledge. JAMA 2022;328(24):2398.
[12] Tamuhla T, Lulamba ET, Mutemaringa T, Tiffin N. Multiple modes of data sharing can facilitate secondary use of sensitive health data for research. BMJ Glob Health 2023;8(10).
[13] Tucker K, Branson J, Dilleen M, Hollis S, Loughlin P, Nixon MJ, Williams Z. Protecting patient privacy when sharing patient-level data from clinical trials. BMC Med Res Methodol 2016;16(S1).
[14] He Y, Huang F, Jiang X, Nie Y, Wang M, Wang J, Chen H. Foundation model for advancing healthcare: challenges, opportunities and future directions. IEEE Rev Biomed Eng 2024;18:172–91.
[15] Johnson KB, Wei W-Q, Weeraratne D, Frisse ME, Misulis K, Rhee K, Zhao J, Snowdon JL. Precision medicine, AI, and the future of personalized health care. Clin Transl Sci 2020;14(1):86–93.
[16] Alowais SA, Alghamdi SS, Alsuhebany N, Alqahtani T, Alshaya AI, Almohareb SN, Aldairem A, Alrashed M, Bin Saleh K, Badreldin HA, et al. Revolutionizing healthcare: The role of artificial intelligence in clinical practice. BMC Med Educ 2023;23(1).
[17] Moor M, Banerjee O, Abad ZSH, Krumholz HM, Leskovec J, Topol EJ, Rajpurkar P. Foundation models for generalist medical artificial intelligence. Nature 2023;616(7956):259–65.
[18] Li C, Wong C, Zhang S, Usuyama N, Liu H, Yang J, Naumann T, Poon H, Gao J. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. Adv Neural Inf Process Syst 2023;36:28541–64.
[19] Acosta JN, Falcone GJ, Rajpurkar P, Topol EJ. Multimodal biomedical AI. Nature Med 2022;28(9):1773–84.
[20] Tu T, Azizi S, Driess D, Schaekermann M, Amin M, Chang P-C, Carroll A, Lau C, Tanno R, Ktena I, et al. Towards generalist biomedical AI. Nejm Ai 2024;1(3). AIoa2300138.
[21] of Canada P.
[22] McMahan B, Moore E, Ramage D, Hampson S, y Arcas BA. Communication-efficient learning of deep networks from decentralized data. In: Artificial intelligence and statistics. PMLR; 2017, p. 1273–82.
[23] Yin Z, Wang H, Chen B, Zhang X, Lin X, Sun H, Li A, Zhou C. Federated semi-supervised representation augmentation with cross-institutional knowledge transfer for healthcare collaboration. Knowl-Based Syst 2024;300:112208.
[24] Wang H, Yin Z, Chen B, Zeng Y, Yan X, Zhou C, Li A. Rofed-llm: robust federated learning for large language models in adversarial wireless environments. IEEE Trans Netw Sci Eng 2025.
[25] Dwork C, Roth A, et al. The algorithmic foundations of differential privacy. Found Trends® Theor Computer Sci 2014;9(3–4):211–407.
[26] Cramer R, Damgård IB, et al. Secure multiparty computation. Cambridge University Press; 2015.
[27] Gentry C. Fully homomorphic encryption using ideal lattices. In: Proceedings of the forty-first annual ACM symposium on theory of computing. 2009, p. 169–78.
[28] Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih W-t, Rocktäschel T, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. Adv Neural Inf Process Syst 2020;33:9459–74.
[29] Li Y, Du X, Wang Y, Chen X, Zhou Z, Lian J, Chuang Y-W, Hong P, Hou PC, Zhou L. Ai-assisted literature screening: A hybrid approach using large language models and retrieval-augmented generation. Int J Med Informatics 2025;106205.
[30] Government of Canada. A Data Strategy Roadmap for the Federal Public Service: Report to the clerk of the privy council. Tech. Rep.,, Ottawa, ON: Privy Council Office; 2018, URL https://www.canada.ca/en/privy-council/corporate/clerk/publications/data-strategy.html.
[31] Guo Y, Hao Z, Zhao S, Gong J, Yang F. Artificial intelligence in health care: bibliometric analysis. J Med Internet Res 2020;22(7):e18228.
[32] Touvron H, Lavril T, Izacard G, Martinet X, Lachaux M-A, Lacroix T, Rozière B, Goyal N, Hambro E, Azhar F, Rodriguez A, Joulin A, Grave E, Lample G. LLaMA: Open and efficient foundation language models. 2023, arXiv preprint.arXiv:2302.13971.
[33] Chase H. Langchain: Building applications with llms through composability. 2022, URL https://github.com/langchain-ai/langchain.
[34] Johnson AE, Bulgarelli L, Shen L, Gayles A, Shammout A, Horng S, Pollard TJ, Hao S, Moody B, Gow B, et al. Mimic-IV, a freely accessible electronic health record dataset. Sci Data 2023;10(1).
[35] Liu H, Li C, Wu Q, Lee YJ. Visual instruction tuning. In: NeurIPS. 2023.
[36] Corbin J, Strauss A. Basics of qualitative research: Techniques and procedures for developing grounded theory. Sage publications; 2014.
[37] Chun Tie Y, Birks M, Francis K. Grounded theory research: A design framework for novice researchers. SAGE Open Med 2019;7:2050312118822927.
[38] Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, Moher D, Peters MD, Horsley T, Weeks L, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 2018;169(7):467–73.
[39] Peters MD, Godfrey CM, Khalil H, McInerney P, Parker D, Soares CB. Guidance for conducting systematic scoping reviews. Int J Evidence-Based Heal 2015;13(3):141–6.
[40] Arksey H, O'Malley L. Scoping studies: towards a methodological framework. Int J Soc Res Methodol 2005;8(1):19–32.
[41] Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. Implement Sci 2010;5(1):69.
[42] Public Health Agency of Canada, Pan-Canadian Health Data Strategy Expert Advisory Group. Expert advisory group report 3: Toward a world-class health data system. 2022, [Accessed 7 June 2025].
[43] Heo S, Chan AY, Diaz Peralta P, Jin L, Pereira Nunes CR, Bell ML. Impacts of the COVID-19 pandemic on scientists' productivity in science, technology, engineering, mathematics (STEM), and medicine fields. Humanit Soc Sci Commun 2022;9(1):1–11.
[44] of the United States Trade Representative O. United States–Mexico–Canada agreement (USMCA), chapter 19: Digital trade. 2020, Articles 19.11 (Cross-Border Transfer of Information) and 19.12 (Location of Computing Facilities). URL https://ustr.gov/sites/default/files/files/agreements/FTA/USMCA/Text/19-Digital-Trade.pdf.
[45] OGL. G7 International Patient Summary Roadmap. Tech. Rep., UK Department of Health and Social Care; 2021, URL https://assets.publishing.service.gov.uk/media/61d82fbd8fa8f505893f1c93/G7-international-patient-summary-roadmap.pdf.
[46] OECD. Facilitating the secondary use of health data for public interest purposes across borders. Tech. Rep., Organisation for Economic Co-operation and Development; 2025.
[47] Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D. Grad-cam: Visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE international conference on computer vision. 2017, p. 618–26.
[48] Zhang H, Qie Y. Applying deep learning to medical imaging: a review. Appl Sci 2023;13(18):10521.
[49] Rahnenführer J, De Bin R, Benner A, Ambrogi F, Lusa L, Boulesteix A-L, Migliavacca E, Binder H, Michiels S, Sauerbrei W, et al. Statistical analysis of high-dimensional biomedical data: a gentle introduction to analytical goals, common approaches and challenges. BMC Med 2023;21(1):182.

[50] Berisha V, Krantsevich C, Hahn PR, Hahn S, Dasarathy G, Turaga P, Liss J. Digital medicine and the curse of dimensionality. NPJ Digit Med 2021;4(1):153.

[51] Ibrahim M, Khalil YA, Amirrajab S, Suna C, Breeuwer M, Pluim J, Elen B, Ertaylan G, Dumontiera M. Generative AI for synthetic data across multiple medical modalities: A systematic review of recent developments and challenges. 2024, arXiv preprint arXiv:2407.00116.

[52] Kush RD, Warzel D, Kush MA, Sherman A, Navarro EA, Fitzmartin R, Pétavy F, Galvez J, Becnel LB, Zhou F, et al. FAIR data sharing: the roles of common data elements and harmonization. J Biomed Inform 2020;107:103421.

[53] Friesen P, Douglas-Jones R, Marks M, Pierce R, Fletcher K, Mishra A, Lorimer J, Véliz C, Hallowell N, Graham M, et al. Governing AI-driven health research: Are IRBs up to the task? Ethics & Hum Res 2021;43(2):35–42.

[54] Althnian A, AlSaeed D, Al-Baity H, Samha A, Dris AB, Alzakari N, Abou Elwafa A, Kurdi H. Impact of dataset size on classification performance: an empirical evaluation in the medical domain. Appl Sci 2021;11(2):796.

[55] Decherchi S, Pedrini E, Mordenti M, Cavalli A, Sangiorgi L. Opportunities and challenges for machine learning in rare diseases. Front Med 2021;8:747612.

[56] Collins GS, Reitsma JB, Altman DG, Moons KG. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. J Br Surg 2015;102(3):148–58.

[57] Christodoulou E, Ma J, Collins GS, Steyerberg EW, Verbakel JY, Van Calster B. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. J Clin Epidemiol 2019;110:12–22.

[58] Yang Q, Liu Y, Chen T, Tong Y. Federated machine learning: Concept and applications. ACM Trans Intell Syst Technol (TIST) 2019;10(2):1–19.

[59] Infoway CH. Shared Pan-Canadian Interoperability Roadmap. Tech. Rep, Canada Health Infoway; 2023.

[60] Alami H, Lehoux P, Papoutsi C, Shaw SE, Fleet R, Fortin J-P. Understanding the integration of artificial intelligence in healthcare organisations and systems through the NASSS framework: a qualitative study in a leading Canadian academic centre. BMC Health Serv Res 2024;24(1):701.

[61] Turnbull C, Scott RH, Thomas E, Jones L, Murugaesu N, Pretty FB, Halai D, Baple E, Craig C, Hamblin A, et al. The 100 000 genomes project: bringing whole genome sequencing to the NHS. Bmj 2018;361.

[62] Sosinsky A, Ambrose J, Cross W, Turnbull C, Henderson S, Jones L, Hamblin A, Arumugam P, Chan G, Chubb D, et al. Insights for precision oncology from the integration of genomic and clinical data of 13,880 tumors from the 100,000 genomes cancer programme. Nature Med 2024;30(1):279–89.

[63] 100 GPPI. 100,000 genomes pilot on rare-disease diagnosis in health care—preliminary report. N Engl J Med 2021;385(20):1868–80.

[64] Mayo KR, Basford MA, Carroll RJ, Dillon M, Fullen H, Leung J, Master H, Rura S, Sulieman L, Kennedy N, et al. The all of us data and research center: creating a secure, scalable, and sustainable ecosystem for biomedical research. Annu Rev Biomed Data Sci 2023;6(1):443–64.

[65] Sudlow C, Gallacher J, Allen N, Beral V, Burton P, Danesh J, Downey P, Elliott P, Green J, Landray M, et al. UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. PLoS Med 2015;12(3):e1001779.

[66] Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, Scales N, Tanwani A, Cole-Lewis H, Pfohl S, et al. Large language models encode clinical knowledge. Nature 2023;620(7972):172–80.

[67] Yang X, Chen A, PourNejatian N, Shin HC, Smith KE, Parisien C, Compas C, Martin C, Flores MG, Zhang Y, et al. Gatortron: A large clinical language model to unlock patient information from unstructured electronic health records. 2022, arXiv preprint arXiv:2203.03540.

[68] Ahmed F, Sellergren A, Yang L, Xu S, Babenko B, Ward A, Olson N, Mohtashamian A, Matias Y, Corrado GS, et al. Pathalign: A vision-language model for whole slide images in histopathology. 2024, arXiv preprint arXiv:2406.19578.

[69] Pellegrini C, Özsoy E, Busam B, Wiestler B, Navab N, Keicher M. Radialog: Large vision-language models for X-Ray reporting and dialog-driven assistance. In: Medical imaging with deep learning. 2025.

[70] Murphy SN, Weber G, Mendis M, Gainer V, Chueh HC, Churchill S, Kohane I. Serving the enterprise and beyond with informatics for integrating biology and the bedside (i2b2). J Am Med Informatics Assoc 2010;17(2):124–30.

[71] Verma AA, Guo Y, Kwan JL, Lapointe-Shaw L, Rawal S, Tang T, Weinerman A, Cram P, Dhalla IA, Hwang SW, et al. Patient characteristics, resource use and outcomes associated with general internal medicine hospital care: the general medicine inpatient initiative (GEMINI) retrospective cohort study. Can Med Assoc Open Access J 2017;5(4):E842–9.

[72] Sweeney L. Only you, your doctor, and many others may know. Technol Sci 2015;2015092903(9):29.

[73] Goncalves A, Ray P, Soper B, Stevens J, Coyle L, Sales AP. Generation and evaluation of synthetic patient data. BMC Med Res Methodol 2020;20:1–40.

[74] Qi Y, Vianna P, Cadrin-Chênevert A, Blanchet K, Montagnon E, Belilovsky E, Wolf G, Mullie L-A, Cloutier G, Chassé M, Tang A. Simulating federated learning for steatosis detection using ultrasound images. Sci Rep 2024;14(1):13253.

[75] Fang C, Dziedzic A, Zhang L, Oliva L, Verma A, Razak F, Papernot N, Wang B. Decentralised, collaborative, and privacy-preserving machine learning for multi-hospital data. EBioMedicine 2024;101.

[76] Roth HR, Cheng Y, Wen Y, Yang I, Xu Z, Hsieh Y-T, Kersten K, Harouni A, Zhao C, Lu K, et al. Nvidia flare: Federated learning from simulation to real-world. 2022, arXiv preprint arXiv:2210.13291.

[77] Tavakoli N, Shakeri Z, Gowda V, Samsel K, Bedayat A, Ghasemiesfe A, Bagci U, Hsiao A, Leiner T, Carr J, et al. Generative AI and foundation models in radiology: Applications, opportunities, and potential challenges. Radiology 2025;317(2):e242961.

[78] Connected care for Canadians act. 2024, URL http://bit.ly/3xfeFkB.

[79] Chomutare T, Tejedor M, Svenning TO, Marco-Ruiz L, Tayefi M, Lind K, Godtliebsen F, Moen A, Ismail L, Makhlysheva A, et al. Artificial intelligence implementation in healthcare: a theory-based scoping review of barriers and facilitators. Int J Environ Res Public Health 2022;19(23):16359.

[80] Nair M, Svedberg P, Larsson I, Nygren JM. A comprehensive overview of barriers and strategies for ai implementation in healthcare: Mixed-method design. PLos One 2024;19(8):e0305949.

[81] Kelly CJ, Karthikesalingam A, Suleyman M, Corrado G, King D. Key challenges for delivering clinical impact with artificial intelligence. BMC Med 2019;17:1–9.